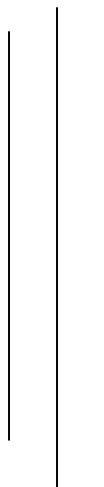
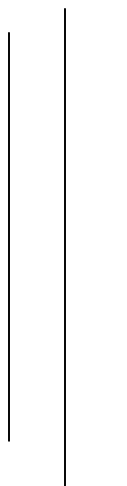


Matrices



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Vectors:

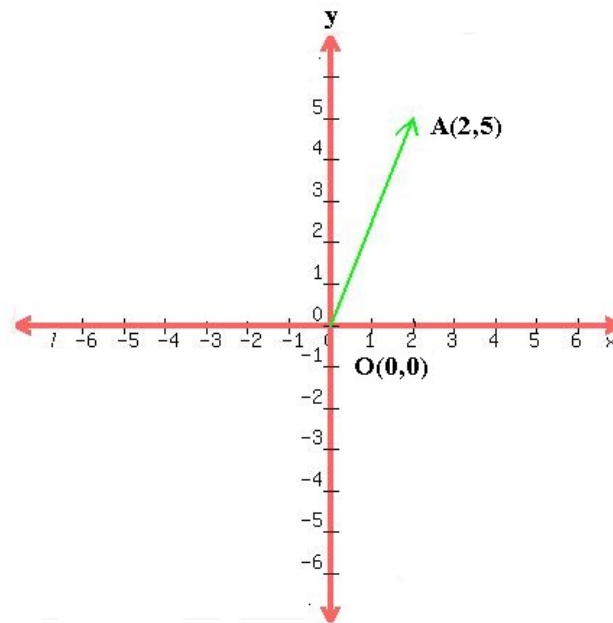


Figure no 01

The Vector $\vec{OA}$ can be written as $= [2,5]$
 $= \begin{pmatrix} 2 \\ 5 \end{pmatrix} \rightarrow$ In the form of Matrix
 $= 2\hat{i} + 5\hat{j}$

A vector can be specified by its components:

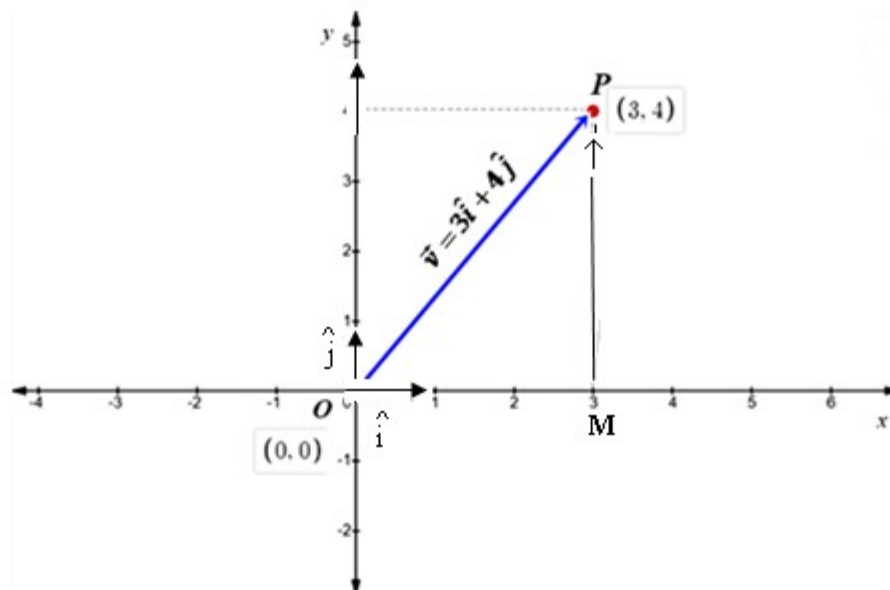


Figure no 02

A vector (Figure no 02) can be expressed as a column matrix:

From, ΔOPM

$$\vec{OP} = \vec{OM} + \vec{MP} = 3\hat{i} + 4\hat{j}$$

$$\text{Or, } \vec{OP} = [3, 4]$$

$$\text{Or, } \vec{OP} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

↑

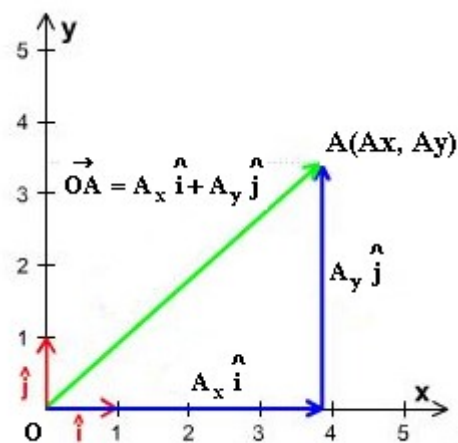


Figure 03

A vector (Figure no 03): 2-dimensional (2-components)

$$\vec{OA} = [A_x, A_y]$$

$$\text{Or, } \vec{OA} = \begin{bmatrix} A_x \\ A_y \end{bmatrix}$$

$$\text{Or, } \vec{OA} = A_x \hat{i} + A_y \hat{j}$$

A vector **Figure no 04: 3-dimensional (3-components)**

$$\vec{a} = [a_x, a_y, a_z]$$

$$\text{Or, } \vec{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}$$

$$\text{Or, } \vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

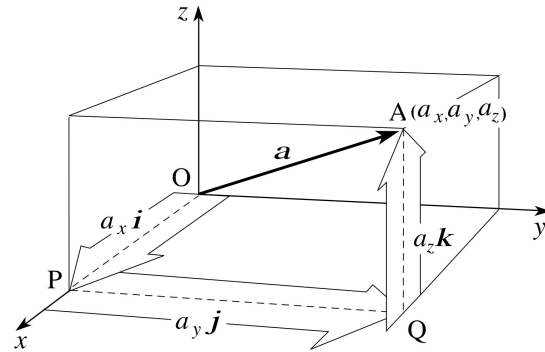


Figure no 04

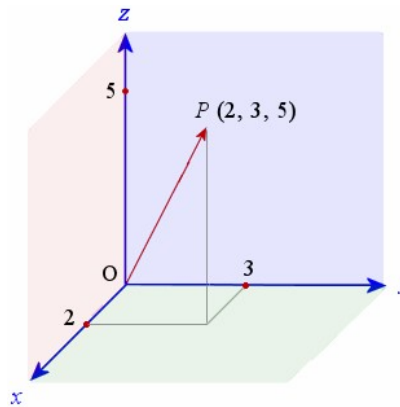


Figure no 05

The numbers comprising the vector are now called *components*, and the number of components equals the dimensionality of the vector. **$\vec{x}$ is a 5-dimensional vector (5-components).**

$$\vec{x} = \begin{pmatrix} 8 \\ 6 \\ 7 \\ 5 \\ 3 \end{pmatrix}$$

Vectors can be equivalently represented horizontally to save space, e.g. $\vec{x} = [8, 6, 7, 5, 3]$ is the same vector as above.

Determinant: A determinant is a function of a square matrix that reduces it to a single number. For example the determinant of the matrix $A = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$ is

$$|A| = \begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix} = 4 \times 2 - 1 \times 1 = 7$$

Matrices: A matrix is probably most familiar as a table of data, like Table 1.

Contestant	Round 1	Round 2	Round 3	Round 4	Total	Place
Carol Semple-Marzetta	17	18	5	5	45	1
Susan Curry	42	28	30	15	115	3
Monica Brant	10	10	10	21	51	2
Karen Hulse	28	5	65	39	132	5
Dale Tomita	24	26	45	21	116	4

Written as a matrix, Table 1 looks like this:

$$\begin{bmatrix} 17 & 18 & 5 & 5 & 45 & 1 \\ 42 & 28 & 30 & 15 & 115 & 3 \\ 10 & 10 & 10 & 21 & 51 & 2 \\ 28 & 5 & 65 & 39 & 132 & 5 \\ 24 & 26 & 45 & 21 & 116 & 4 \end{bmatrix}$$

The maximum number of rows is assigned to the variable m , and the number of columns is called n . Matrix *entries* (also called *elements* or *components*) are denoted by a lower-case a , and a *particular entry is referenced by its row index (labeled i) and its column index (labeled j)*.

Matrix Notation

A matrix is basically a collection of vectors. We can talk about row vectors or column vectors. A vector can be expressed as a column matrix:

$$\begin{aligned} \vec{v} &= [v_x, v_y, v_z] \\ \Rightarrow \vec{v} &= \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \end{aligned}$$

A system of any $m \times n$ numbers arranged in a rectangular array of ‘ m ’ rows and ‘ n ’ columns is called a Matrix of order $m \times n$ or an $n \times m$ Matrix which is read as m by n or n by m Matrix. Any rectangular arrangement or any array of numbers enclosed in brackets such as $[]$, $()$, $\| \|$

Matrix can be denoted by $[]$, $()$, $\| \|$

Example:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} & - & - & - & - & - & a_{1n} \\ a_{21} & a_{22} & a_{23} & - & - & - & - & - & a_{2n} \\ a_{31} & a_{32} & a_{33} & - & - & - & - & - & a_{3n} \\ - & - & - & - & - & - & - & - & - \\ a_{m1} & a_{m2} & a_{m3} & - & - & - & - & - & a_{mn} \end{pmatrix}$$

Where $i = 1, 2, 3, 4, \dots, m$ and $j = 1, 2, 3, 4, \dots, n$

Where $i \rightarrow \text{row}, j \rightarrow \text{column}$

[Here a_{11} means 1st row 1st column, a_{12} means 1st row 2nd column.....]

For example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \text{ is a } 2 \times 3 \text{ order matrix}$$

Magnitude of a vector: The magnitude of a vector $\vec{OP}$ or $\vec{V}$ is its length and is normally denoted by $|\vec{V}|$ or V . Given a vector $\vec{V}$ with tail at the origin O and head at $P(x, y)$, what's its length?

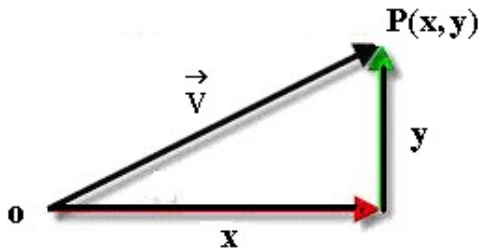


Figure: 06

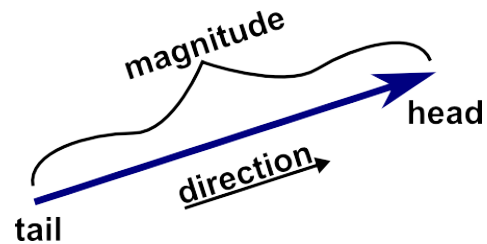


Figure: 07

According to Pythagoras, the length of the hypotenuse OP is the square root of $x^2 + y^2$.

That is, Magnitude of a vector $\vec{V} = \text{its length} = |\vec{OP}| = |\vec{V}| = \sqrt{x^2 + y^2}$

If $\vec{u}$ is a vector, its length is denoted by $|\vec{u}|$ more concisely,

$$|\vec{u}| = \sqrt{\sum_{i=1}^n x_i^2}$$

For example, if $\vec{u} = [4, 11, 8, 10]$ then

$$|\vec{u}| = \sqrt{4^2 + 11^2 + 8^2 + 10^2} = \sqrt{301} = 17.35$$

Vector Addition:

Adding two vectors means adding each component in $\vec{u}_1$ to the component in the corresponding position in $\vec{u}_2$ to get a new vector.

For example:

$$\vec{u}_1 = [3, 2, 1, -2] \text{ and } \vec{u}_2 = [2, -1, 4, 1]$$

$$\vec{u}_1 + \vec{u}_2 = [3, 2, 1, -2] + [2, -1, 4, 1] = [(3+2), (2-1), (1+4), (-2+1)] = [5, 1, 5, -1]$$

Scalar Multiplication:

Multiplying a scalar (real number) times a vector means multiplying every component by that real number to yield a new vector.

$$\text{For instance, if } \vec{u} = [3, 6, 8, 4] \text{ then } 1.5 * \vec{u} = 1.5 * [3, 6, 8, 4] = [4.5, 9, 12, 6]$$

More generally, *scalar multiplication* means if d is a real number and $\vec{u}$ is a vector $[u_1, u_2, \dots, u_n]$, then $d * \vec{u} = [du_1, du_2, \dots, du_n]$

Inner Product

The *inner product* of two vectors (also called the *dot product* or *scalar product*) defines multiplication of vectors. The inner product is only defined for vectors of the same dimension. The inner product of two vectors is denoted $(\vec{u}_1, \vec{u}_2)$ or $\vec{u}_1 \cdot \vec{u}_2$ (the dot product). Thus, $(\vec{x}, \vec{y}) = \vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i$

For example, if $\vec{x} = [1, 6, 7, 4]$ and $\vec{y} = [3, 2, 8, 3]$, then

$$\vec{x} \cdot \vec{y} = 1(3) + 6(2) + 7(8) + 3(4) = 83$$

Orthogonality

Two vectors are orthogonal to each other if their inner product equals zero. In two-dimensional space this is equivalent to saying that the vectors are perpendicular, or that the only angle between them is a 90° angle. For example, the vectors $[2, 1, -2, 4]$ and $[3, -6, 4, 2]$ are orthogonal because

$$[2, 1, -2, 4] \cdot [3, -6, 4, 2] = 2(3) + 1(-6) - 2(4) + 4(2) = 0$$

Normal Vector

A normal vector (or unit vector) is a vector of length 1. Any vector with an initial length > 0 , can be normalized by dividing each component in it by the vectors length.

For example, if $\vec{v} = [2, 4, 1, 2]$ then

$$\left| \vec{v} \right| = \sqrt{2^2 + 4^2 + 1^2 + 2^2} = \sqrt{25} = 5 \text{ is not a normal vector}$$

But $\vec{u} = [\frac{2}{5}, \frac{4}{5}, \frac{1}{5}, \frac{2}{5}]$ is a normal vector because

$$|\vec{u}| = \sqrt{(2/5)^2 + (4/5)^2 + (1/5)^2 + (2/5)^2} = \sqrt{25/25} = 1$$

Orthonormal Vectors

Vectors of unit length that are orthogonal to each other are said to be *orthonormal*. For example,

$$\vec{u} = [2/5, 1/5, -2/5, 4/5]$$

and $\vec{v} = [3/\sqrt{65}, -6/\sqrt{65}, 4/\sqrt{65}, 2/\sqrt{65}]$

are orthonormal because

$$|\vec{u}| = \sqrt{(2/5)^2 + (1/5)^2 + (-2/5)^2 + (4/5)^2} = 1$$

$$|\vec{v}| = \sqrt{(3/\sqrt{65})^2 + (-6/\sqrt{65})^2 + (4/\sqrt{65})^2 + (2/\sqrt{65})^2} = 1$$

$$\vec{u} \cdot \vec{v} = \frac{6}{5\sqrt{65}} - \frac{6}{5\sqrt{65}} - \frac{8}{5\sqrt{65}} + \frac{8}{5\sqrt{65}} = 0$$

What does a pair of orthonormal vectors in 2-D Euclidean space look like?

Let $\mathbf{u} = (x_1, y_1)$ and $\mathbf{v} = (x_2, y_2)$. Consider the restrictions on x_1, x_2, y_1, y_2 required to make $\mathbf{u}$ and $\mathbf{v}$ form an orthonormal pair.

- From the orthogonality restriction, $\mathbf{u} \cdot \mathbf{v} = 0$.
- From the unit length restriction on $\mathbf{u}$, $\|\mathbf{u}\| = 1$.
- From the unit length restriction on $\mathbf{v}$, $\|\mathbf{v}\| = 1$.

Expanding these terms gives 3 equations:

- $x_1x_2 + y_1y_2 = 0$
- $\sqrt{x_1^2 + y_1^2} = 1$
- $\sqrt{x_2^2 + y_2^2} = 1$

Gram-Schmidt Orthonormalization Process

The Gram-Schmidt orthonormalization process is a method for converting a set of vectors into a set of orthonormal vectors. It basically begins by normalizing the first vector under consideration and iteratively rewriting the remaining vectors in terms of themselves minus a multiplication of the already normalized vectors. For example, to convert the column vectors of

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 0 \\ 2 & 3 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

into orthonormal column vectors

$$A = \begin{bmatrix} \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{6} & \frac{2}{3} \\ 0 & \frac{2\sqrt{2}}{3} & \frac{-1}{3} \\ \frac{\sqrt{6}}{3} & 0 & 0 \\ \frac{\sqrt{6}}{6} & \frac{-\sqrt{2}}{6} & \frac{-2}{3} \end{bmatrix},$$

First normalize $\vec{v}_1 = [1, 0, 2, 1]$

$$\vec{u}_1 = \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right]$$

Next, let

$$\begin{aligned} \vec{w}_2 &= \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 = [2, 2, 3, 1] - \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right] \cdot [2, 2, 3, 1] * \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right] \\ &= [2, 2, 3, 1] - \left(\frac{9}{\sqrt{6}} \right) * \left[\frac{1}{\sqrt{6}}, 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right] \\ &= [2, 2, 3, 1] - \left[\frac{3}{2}, 0, 3, \frac{3}{2} \right] \\ &= \left[\frac{1}{2}, 2, 0, \frac{-1}{2} \right] \end{aligned}$$

Normalize $\vec{w}_2$ to get $\vec{u}_2 = \left[\frac{\sqrt{2}}{6}, \frac{2\sqrt{2}}{3}, 0, \frac{-\sqrt{2}}{6} \right]$

Now compute $\vec{u}_3$ in terms of $\vec{u}_1$ and $\vec{u}_2$ as follows. Let

$$\vec{w}_3 = \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 - \vec{u}_2 \cdot \vec{v}_2 * \vec{u}_2 = \left[\frac{4}{9}, \frac{-2}{9}, 0, \frac{-4}{9} \right]$$

and normalize $\vec{w}_3$ to get $\vec{u}_3 = \left[\frac{2}{3}, \frac{-1}{3}, 0, \frac{-2}{3} \right]$

More generally, if we have an orthonormal set of vectors $\vec{u}_1, \dots, \vec{u}_{k-1}$, then $\vec{w}_k$ is

$$\text{expressed as } \vec{w}_k = \vec{v}_k - \sum_{i=1}^{k-1} \vec{u}_i \cdot \vec{v}_k * \vec{u}_i$$

We define the projection operator by

$$\text{Proj}_{\mathbf{u}} \mathbf{v} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u} = \langle \mathbf{u}, \mathbf{v} \rangle \frac{\mathbf{u}}{\langle \mathbf{u}, \mathbf{u} \rangle},$$

Where $\langle \mathbf{u}, \mathbf{v} \rangle$ denotes the **inner product** of the vectors $\mathbf{u}$ and $\mathbf{v}$. This operator projects the vector $\mathbf{v}$ orthogonally onto the vector $\mathbf{u}$.

The Gram–Schmidt process then works as follows:

$$\begin{aligned} \mathbf{u}_1 &= \mathbf{v}_1, & \mathbf{e}_1 &= \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} \\ \mathbf{u}_2 &= \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_2, & \mathbf{e}_2 &= \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} \\ \mathbf{u}_3 &= \mathbf{v}_3 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_3 - \text{proj}_{\mathbf{u}_2} \mathbf{v}_3, & \mathbf{e}_3 &= \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} \\ \mathbf{u}_4 &= \mathbf{v}_4 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_4 - \text{proj}_{\mathbf{u}_2} \mathbf{v}_4 - \text{proj}_{\mathbf{u}_3} \mathbf{v}_4, & \mathbf{e}_4 &= \frac{\mathbf{u}_4}{\|\mathbf{u}_4\|} \\ &\vdots & &\vdots \\ \mathbf{u}_k &= \mathbf{v}_k - \sum_{j=1}^{k-1} \text{proj}_{\mathbf{u}_j} \mathbf{v}_k, & \mathbf{e}_k &= \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|} \end{aligned}$$

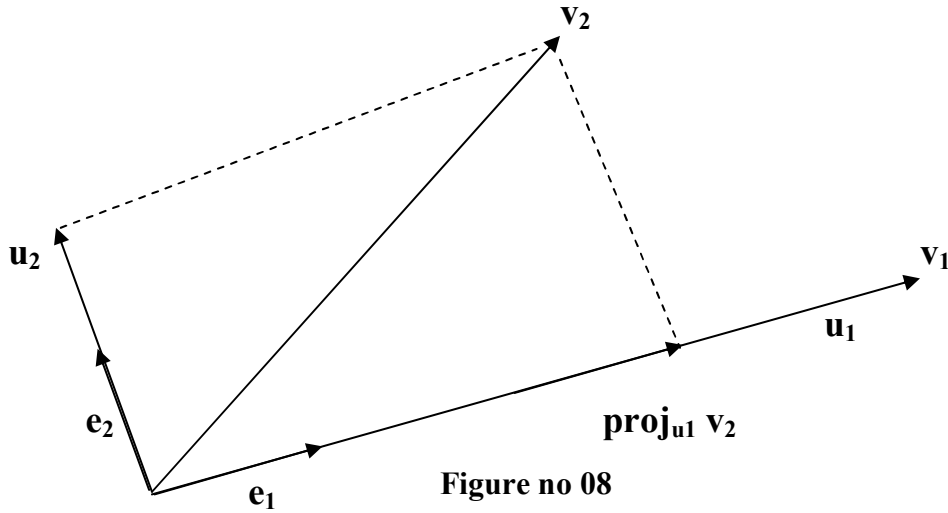


Figure no 08

The sequence $\mathbf{u}_1, \dots, \mathbf{u}_k$ is the required system of orthogonal vectors, and the normalized vectors $\mathbf{e}_1, \dots, \mathbf{e}_k$ form an *orthonormal set*.

To check that these formulas yield an orthogonal sequence, first compute $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle$ by substituting the above formula for $\mathbf{u}_2$: you will get zero. Then use this to compute $\langle \mathbf{u}_1, \mathbf{u}_3 \rangle$ again by substituting the formula for $\mathbf{u}_3$: you will get zero. The general proof proceeds by mathematical induction.

Geometrically, this method proceeds as follows: to compute $\mathbf{u}_i$, it projects $\mathbf{v}_i$ orthogonally onto the subspace U generated by $\mathbf{u}_1, \dots, \mathbf{u}_{i-1}$, which is the same as the subspace generated by $\mathbf{v}_1, \dots, \mathbf{v}_{i-1}$. The vector $\mathbf{u}_i$ is then defined to be the difference between $\mathbf{v}_i$ and this projection, guaranteed to be orthogonal to all of the vectors in the subspace U .

The Gram–Schmidt process also applies to a linearly independent infinite sequence $\{\mathbf{v}_i\}_i$. The result is an orthogonal (or orthonormal) sequence $\{\mathbf{u}_i\}_i$ such that for natural number n : the algebraic span of $\mathbf{v}_1, \dots, \mathbf{v}_n$ is the same as that of $\mathbf{u}_1, \dots, \mathbf{u}_n$.

If the Gram–Schmidt process is applied to a linearly dependent sequence, it outputs the $\mathbf{0}$ vector on the i th step, assuming that $\mathbf{V}_i$ is a linear combination of $\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_{i-1}$.

Example:

Consider the following set of vectors in $\mathbb{R}^2$ (with the conventional inner product)

$$S = \left\{ \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}.$$

Now, perform Gram-Schmidt, to obtain an orthogonal set of vectors:

$$\mathbf{u}_1 = \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \text{proj}_{\begin{pmatrix} 3 \\ 1 \end{pmatrix}} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix}.$$

We check that the vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ are indeed orthogonal:

$$\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \left\langle \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix} \right\rangle = -\frac{6}{5} + \frac{6}{5} = 0.$$

We can then normalize the vectors by dividing out their sizes as shown above:

$$\mathbf{e}_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\mathbf{e}_2 = \frac{1}{\sqrt{\frac{40}{25}}} \begin{pmatrix} -2/5 \\ 6/5 \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

Distinguish between Determinant and Matrices

Determinant	Matrices
1. Determinant can be denoted by: $\begin{vmatrix} & \\ & \end{vmatrix}$ 2. Number of rows and columns are always equal Example: $\begin{vmatrix} 2 & 9 \\ 6 & 1 \end{vmatrix}$ 3. It has a definite value at through it is an arrangement of numbers 4. If two rows (or columns) are identical, a determinant vanishes 5. Rows and columns can be interchanged 6. If a determinant is multiplied by a number m, every element of any one row (or any one column) is multiplied by m. Example: $2 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} 2a & 2b \\ 2c & 2d \end{vmatrix}$ 7. A determinant can be resolved into two determinants by using only one row (or column) 8. The sign of a determinant changes if two rows (or columns) are interchanged 9. The product of two determinants does not change the order Example: $\begin{vmatrix} 1 & 5 \\ 3 & 1 \end{vmatrix} \times \begin{vmatrix} 3 & 1 \\ 7 & 5 \end{vmatrix}$ $= \begin{vmatrix} 1 \times 3 + 5 \times 7 & 1 \times 1 + 5 \times 5 \\ 3 \times 3 + 1 \times 7 & 3 \times 1 + 1 \times 5 \end{vmatrix}$ $= \begin{vmatrix} 38 & 26 \\ 16 & 8 \end{vmatrix}$	1. Matrix can be denoted by: $[]$, $()$, $\ \$ 2. Number of rows and columns are not always equal Example: $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ 3. It has no definite value, it is merely an arrangement of numbers 4. Identical rows (or columns) may occur in a matrix 5. Rows and columns cannot be interchanged 6. If a matrix is multiplied by m, every element of the entire matrix is multiplied by m Example: $2 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} 2a & 2b \\ 2c & 2d \end{vmatrix}$ 7. A matrix cannot be resolved into two matrices using only one row (or column) 8. Two rows (or columns) cannot be interchanged at all 9. The product of two matrices may change the order Example: $(2 \ 4)_{1 \times 2} \times \begin{pmatrix} 5 \\ 8 \end{pmatrix}_{2 \times 1} = (2 \times 5 + 4 \times 8) = (42)_{1 \times 1}$

Square Matrix:

If $m = n$ that is the number of rows & columns of matrix are equal then the matrix is of order $n \times n$ or $m \times m$ is called a square matrix of order n or m .

For example:

$$\begin{bmatrix} 1 & 2 & 6 \\ 2 & 4 & 7 \\ 3 & 5 & 8 \end{bmatrix} \text{ is a } 3 \times 3 \text{ order matrix}$$

Rectangular Matrix:

The number of rows & columns of a matrix need not be equal that is $m \neq n$ then the matrix is known as a rectangular matrix.

For example:

$$\begin{bmatrix} 1 & 4 & 7 & 1 \\ 2 & 5 & 8 & 2 \\ 3 & 6 & 9 & 3 \end{bmatrix} \text{ is a } 3 \times 4 \text{ order matrix}$$

Horizontal Matrix:

If in a matrix the number of columns is more than the number of rows then it is called the horizontal matrix.

For example:

$$\begin{bmatrix} 1 & 4 & 7 & 1 \\ 2 & 5 & 8 & 2 \\ 6 & 5 & 4 & 9 \end{bmatrix} \text{ is a } 3 \times 4 \text{ order matrix}$$

Vertical Matrix:

If in a matrix the number of rows is more than the number of columns then it is called the vertical matrix.

For example:

$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \text{ is a } 3 \times 2 \text{ order matrix}$$

Row Matrix:

If in a matrix there is only one row it is called a row matrix.

For example:

$$[1 \quad 2 \quad 3 \quad 4]$$

Column Matrix:

If in a matrix there is only one column it is called a column matrix.

For example:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Null or Zero Matrix:

If all the elements of an $m \times n$ matrix are zero then it is called a null or zero matrix. It is denoted by 0.

For example:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ or simply } 0$$

Unit Matrix/Identity Matrix:

A square matrix having unity for its elements in the leading diagonal and all other elements as zero is called a unit matrix.

For example:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is a } 3 \times 3 \text{ order matrix}$$

Diagonal Matrix:

A square matrix in which all elements except those in the main or leading diagonal are zero is known as a diagonal matrix.

For example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \text{ is a } 3 \times 3 \text{ order matrix}$$

Trace of a matrix

The sum of the diagonal elements of a square matrix is called the trace of a matrix.

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 2 & 1 \\ 2 & 9 & 3 \end{bmatrix}$$

Trace of a matrix : $1 + 2 + 3 = 6$

Tridiagonal matrix

A tridiagonal matrix has nonzero elements only on the main diagonal, the diagonal upon the main diagonal, and the diagonal below the main diagonal. This special structure appears often in scientific computing and computer graphics

$$\begin{pmatrix} 1 & 4 & 0 & 0 \\ 3 & 4 & 1 & 0 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 3 & 5 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 0 \\ 0 & 3 & 4 & 2 & 0 \\ 0 & 0 & 5 & 1 & 2 \\ 0 & 0 & 0 & 5 & 2 \end{pmatrix}$$

Reciprocal Matrix (Comparison Matrix)

Making Comparison Matrix

In this section you will learn how to make a reciprocal matrix from pair wise comparisons.

For example John has 3 kinds of fruits to be compared and he made subjective judgment on which fruit he likes best, like the following

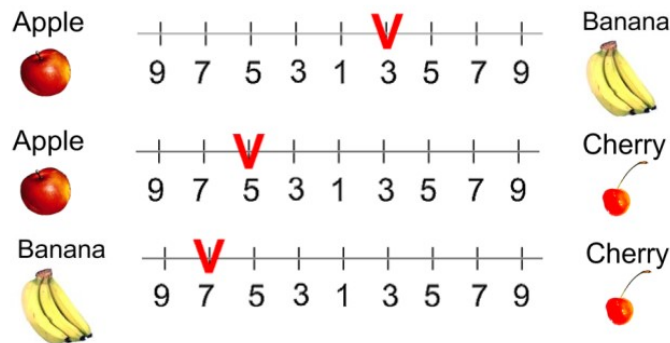


Figure no 09

First he prefers Banana (B) to Apple (A). Thus we say that for John, Banana (B) has greater value than Apple (A). Next, he prefers Apple (A) to Cherry (C). For him, Apple (A) has greater value than Cherry (C). In the last comparison he likes Banana (B) more than Cherry (C). That is, We write it as $B > A$, We write it as $A > C$ Since $B > A$ and $A > C$, logically, we hope that $B > C$.

To fill the lower triangular matrix, we use the reciprocal values of the upper diagonal. If a_{ij} is the element of row i column j of the matrix, then the lower diagonal is filled using this formula:

$$a_{ji} = \frac{1}{a_{ij}}$$

$$D = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{pmatrix} 1 & \frac{1}{3} & 5 \\ 3 & 1 & 7 \\ \frac{1}{5} & \frac{1}{7} & 1 \end{pmatrix} \end{matrix}$$

Decision Matrix

A decision matrix is a tool to evaluate and select the best option between different choices

Example of Normalized Decision Matrix:

We sum each column of the above reciprocal matrix to get

$$A = \begin{pmatrix} 1 & \frac{1}{3} & 5 \\ 3 & 1 & 7 \\ \frac{1}{5} & \frac{1}{7} & 1 \end{pmatrix}$$

$$\text{Sum} \quad \frac{21}{5} \quad \frac{31}{21} \quad 13$$

Then we divide each element of the matrix with the sum of its column, we have normalized relative weight. The sum of each column is 1. The normalized matrix is:

$$A = \begin{pmatrix} \frac{5}{21} & \frac{7}{31} & \frac{5}{13} \\ \frac{15}{21} & \frac{21}{31} & \frac{7}{13} \\ \frac{1}{21} & \frac{3}{31} & \frac{1}{13} \end{pmatrix}$$

$$\text{Sum} \quad 1 \quad 1 \quad 1$$

The normalized principal Eigen vector can be obtained by averaging across the rows

$$w = \frac{1}{3} \begin{pmatrix} \frac{5}{21} + \frac{7}{31} + \frac{5}{13} \\ \frac{15}{21} + \frac{21}{31} + \frac{7}{13} \\ \frac{1}{21} + \frac{3}{31} + \frac{1}{13} \end{pmatrix} = \begin{bmatrix} 0.28 \\ 0.64 \\ 0.07 \end{bmatrix}$$

The normalized principal Eigen vector is also called **priority vector**. Since it is normalized, the sum of all elements in priority vector is 1.

Confusion Matrix

A table showing correct predictions and types of incorrect predictions. The confusion matrix is made up of True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN)

True Positives (TP): The model predicted positive and the actual label is positive [[ট্রু পাওয়ার কথা ছিল সেটা ট্রু পাওয়া গেছে]

True Negative (TN): The model predicted negative and the actual label is negative [[ট্রু পাওয়ার কথা ছিল সেটা ট্রু পাওয়া যায়নি Negative পাওয়া গেছে]

False Positive (FP): The model predicted positive and the actual label was negative [[False পাওয়ার কথা ছিল সেটা False পাওয়া গেছে]

False Negative (FN): The model predicted negative and the actual label was positive [False পাওয়ার কথা ছিল সেটা False পাওয়া যায়নি ভুল পাওয়া গেছে]

		Prediction	
		Cat	Dog
Actual	Cat	15	35
	Dog	40	10

[এখানে 50টি বিড়ালের ছবি এবং 50টি কুকুরের ছবি রয়েছে। যাইহোক, 50টি বিড়ালের ছবির মধ্যে, আমাদের অ্যালগরিদম সঠিকভাবে ভবিষ্যদ্বাণী করেছে যে 15টি বিড়ালের ছবি এবং বাকি 35টি কুকুরের। অন্যদিকে, অ্যালগরিদম 50টি কুকুরের ছবির মধ্যে মাত্র 10টি কুকুর হওয়ার ভবিষ্যদ্বাণী করেছে]

Sub-Matrix:

A matrix, which is obtained from a given matrix by deleting any number of rows and number of columns, is called a sub-matrix of the given matrix.

For example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix} \text{ the Sub - Matrix of } A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

Periodic Matrix: A square matrix A is called periodic, if $A^{k+1} = A$, where k is a positive integer.

Idempotent Matrix: A square matrix A is called idempotent provided it satisfies the relation $A^2 = A$.

Nilpotent Matrix: A square matrix A is called Nilpotent Matrix of order m, provided it satisfies the relation $A^m = 0$ and $A^{m-1} \neq 0$, where m is a positive integer 0 is the null matrix.

Involutory Matrix: A square matrix A is called Involutory Matrix , provided it satisfies the relation $A^2 = I$, where I is the identity matrix.

Example 01: Show that the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent.

Answer: $A^2 = A.A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$

$$\begin{aligned}
A.A &= \begin{bmatrix} 2 \times 2 + (-2) \times (-1) + (-4) \times 1 & 2 \times (-2) + (-2) \times 3 + (-4) \times (-2) & 2 \times (-4) + (-2) \times 4 + (-4) \times (-3) \\ (-1) \times 2 + 3 \times (-1) + 4 \times 1 & (-1) \times (-2) + 3 \times 3 + 4 \times (-2) & (-1) \times (-4) + 3 \times 4 + 4 \times (-3) \\ 1 \times 2 + (-2) \times (-1) + (-3) \times 1 & 1 \times (-2) + (-2) \times 3 + (-3) \times (-2) & 1 \times (-4) + (-2) \times 4 + (-3) \times (-3) \end{bmatrix} \\
&= \begin{bmatrix} 4+2-4 & -4-6+8 & -8-8+12 \\ -2-3+4 & 2+9-8 & 4+12-12 \\ 2+2-3 & -2-6+6 & -4-8+9 \end{bmatrix} \\
&= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A
\end{aligned}$$

Example 02: Show that the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is involutory matrix.

$$\begin{aligned}
\text{Answer: } A^2 &= A.A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\
A^2 &= A.A = \begin{pmatrix} 1 \times 1 + 0 \times 0 & 1 \times 0 + 0 \times (-1) \\ 0 \times 1 + (-1) \times 0 & 0 \times 0 + (-1) \times (-1) \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I
\end{aligned}$$

Triangular Matrix:

There are two types of triangular matrix. They are-

Upper Triangular Matrix:

A square matrix 'A' whose elements $a_{ij} = 0$ for $i > j$ is called an upper triangular matrix.

For example:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} & - & - & - & - & - & a_{1n} \\ 0 & a_{22} & a_{23} & - & - & - & - & - & a_{2n} \\ 0 & 0 & a_{33} & - & - & - & - & - & a_{3n} \\ - & - & - & - & - & - & - & - & - \\ 0 & 0 & 0 & - & - & - & - & - & a_{mn} \end{pmatrix} \quad \text{Where } i \rightarrow \text{row, } j \rightarrow \text{column}$$

Lower Triangular Matrix:

A square matrix A whose elements $a_{ij} = 0$ for $j > i$ is called a lower triangular matrix.

For example:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & 0 & 0 & - & - & - & - & - & 0 \\ a_{21} & a_{22} & 0 & - & - & - & - & - & 0 \\ a_{31} & a_{32} & a_{33} & - & - & - & - & - & 0 \\ - & - & - & - & - & - & - & - & - \\ a_{m1} & a_{m2} & a_{m3} & - & - & - & - & - & a_{mn} \end{pmatrix}$$

Scalar Matrix:

If in a square matrix A all the leading diagonal elements are equal to a where $a \neq 0$ and all the remaining elements are equal to zero then it is called a scalar matrix.

For example:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Singular Matrix: A square matrix that does not have a matrix inverse. *A matrix is singular iff its determinant is 0.* For example, there are 10 singular 2×2 (0, 1)-matrices:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Example 03:

If $\begin{bmatrix} 3 & -1+x & 2 \\ 3 & -1 & x+2 \\ x+3 & -1 & 2 \end{bmatrix}$ is singular and $x \in [-4, -1]$, what is the value of x

Binary matrix: A (0, 1)-integer matrix, i.e., a matrix each of whose elements is 0 or 1, also called a binary matrix

Example 04: Test the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ is singular or not

Solution:

$$|A| = 1(4 \times 6 - 5 \times 5) - 2(2 \times 6 - 3 \times 5) + 3(2 \times 5 - 3 \times 4)$$

$$|A| = 1(24 - 25) - 2(12 - 15) + 3(10 - 12)$$

$$= 1(-1) - 2(-3) + 3(-2)$$

$$= -1 + 6 - 6$$

$$= -1$$

As $|A| \neq 0$, hence the matrix is not singular.

Complex Matrix: The matrix consists of complex numbers is called complex Matrix.

Example: $A = \begin{bmatrix} 1+i & 2+3i \\ 2 & 3i \end{bmatrix}$

Complex Conjugate or Conjugate Matrix:

The matrix obtained from any given matrix A of order $m \times n$ with complex elements a_{ij} by replacing its elements by the corresponding conjugate complex numbers is called the complex conjugate or conjugate of A denoted by $\bar{A}$ and read as A conjugate. For example:

$$\text{If } A = \begin{bmatrix} 1+i & 2+3i \\ 2 & 3i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1-i & 2-3i \\ 2 & -3i \end{bmatrix}$$

Real Matrix:

A matrix A is called real if it satisfies the relation $A = \bar{A}$.

For example:

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$$

Imaginary Matrix:

A matrix A is called imaginary provided if satisfies the relation $A = -\bar{A}$.

For example:

$$A = \begin{bmatrix} 2i & 0 \\ 4i & 3i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} -2i & 0 \\ -4i & -3i \end{bmatrix}$$

$$-\bar{A} = \begin{bmatrix} 2i & 0 \\ 4i & 3i \end{bmatrix} = A$$

Transpose of a Matrix: The transpose of a matrix is created by converting its rows into columns; that is, row 1 becomes column 1, row 2 becomes column 2 etc. The transpose of matrix A is A^T For example, if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Then its transpose is

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Symmetric Matrix:

A square matrix $A = (a_{ij})$ is called symmetric provided $a_{ij} = a_{ji}$ for all values of i and j . For example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Here $a_{11} = a_{11}, a_{12} = a_{21}, a_{13} = a_{31}, a_{23} = a_{32}, a_{22} = a_{22}, a_{33} = a_{33}$

Another definition: A symmetric matrix is a square matrix that satisfies

$$A^T = A$$

Where A^T denotes the transpose, so $a_{ij} = a_{ji}$.

Example 05:

If the matrix $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric, find the value of x and hence present the matrix A

Skew Symmetric Matrix:

A square matrix $A = (a_{ij})$ is called skew symmetric matrix $a_{ij} = -a_{ji}$ all values of i and j .

For example:

$$A = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$$

Here,

$$a_{ij} = -a_{ji}$$

$$a_{11} = -a_{11} [i = 1, j = 1]$$

$$\Rightarrow 2a_{11} = 0 \therefore a_{11} = 0$$

Again

$$a_{ij} = -a_{ji}$$

$$a_{22} = -a_{22} [i = 2, j = 2]$$

$$\Rightarrow 2a_{22} = 0 \therefore a_{22} = 0$$

Again

$$a_{ij} = -a_{ji}$$

$$a_{33} = -a_{33} [i = 3, j = 3]$$

$$\Rightarrow 2a_{33} = 0 \therefore a_{33} = 0$$

and $a_{12} = -a_{21}, a_{13} = -a_{31}, a_{23} = -a_{32}$

Self-Adjoint Matrix:

A matrix A for which

$$A^H \equiv \overline{A^T} = A$$

Where the conjugate transpose is denoted by A^H , A^T , is the transpose of A , and $\bar{Z}$ is the complex conjugate. If a matrix is self-adjoint, it is said to be Hermitian.

Hermitian Matrix:

A square matrix is called Hermitian if it is self-adjoint. That is it satisfies $A^H \equiv \overline{A^T} = A$.

Example 06: Prove that $A = \begin{pmatrix} 3 & 2+i \\ 2-i & 1 \end{pmatrix}$ is a Hermitian matrix.

Proof:

$$A^T = \begin{pmatrix} 3 & 2-i \\ 2+i & 1 \end{pmatrix}$$

$$\therefore A^H = \overline{A^T} = \begin{pmatrix} 3 & 2+i \\ 2-i & 1 \end{pmatrix} = A$$

Unitary Matrix:

If the conjugate transpose of a square matrix is equal to its inverse, then it is a unitary matrix, that

$$\text{is, } A^{-1} = \overline{A^T} = A^H$$

$$\text{We have, } AA^{-1} = I$$

How to prove a matrix is unitary:

$$A^{-1} = \overline{A^T} = A^H$$

$$\Rightarrow A^{-1} = A^H$$

$$\Rightarrow A^H = A^{-1}$$

$$\Rightarrow AA^H = AA^{-1}$$

$$\Rightarrow AA^H = I$$

Example 07: Show that the matrix $A = \begin{bmatrix} \frac{1}{2} - \frac{i}{2} & \frac{1}{2} + \frac{i}{2} \\ \frac{1}{2} + \frac{i}{2} & \frac{1}{2} - \frac{i}{2} \end{bmatrix}$ is unitary

$$\text{Given, } A = \begin{bmatrix} \frac{1}{2} - \frac{i}{2} & \frac{1}{2} + \frac{i}{2} \\ \frac{1}{2} + \frac{i}{2} & \frac{1}{2} - \frac{i}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1-i}{2} & \frac{1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix}$$

$$\therefore A^T = \frac{1}{2} \begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix}$$

$$\overline{A^T} = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$

$$\text{That is, } A^H = \overline{A^T} = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$

Now, we have to prove, $AA^H = I$

$$= \frac{1}{2} \begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix} \times \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 1^2 - i^2 + 1^2 - i^2 & (1-i)^2 + (1+i)^2 \\ (1+i)^2 + (1-i)^2 & 1^2 - i^2 + 1^2 - i^2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 1+1+1+1 & 1-2i+i^2+1+2i+i^2 \\ 1+2i+i^2+1-2i+i^2 & 1+1+1+1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4 & 1-2i-1+1+2i-1 \\ 1+2i-1+1-2i-1 & 4 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$

Skew-Hermitian Matrix:

A skew Hermitian matrix is a square matrix A if and only if its conjugate transpose is equal to its negative of itself. i.e., $A^H = \overline{A^T} = -A$, where A^H is the conjugate transpose.

Example 08: Find the following matrix is skew-Hermitian:

$$\begin{pmatrix} 5i & 3-2i \\ -3-2i & 5i \end{pmatrix}$$

$$\text{Let } A = \begin{pmatrix} 5i & 3-2i \\ -3-2i & 5i \end{pmatrix}$$

$$\Rightarrow A^T = \begin{pmatrix} 5i & -3-2i \\ 3-2i & 5i \end{pmatrix}$$

$$\Rightarrow \overline{A^T} = \begin{pmatrix} -5i & -3+2i \\ 3+2i & -5i \end{pmatrix} \text{-----(i)}$$

$$\text{Given, } A = \begin{pmatrix} 5i & 3-2i \\ -3-2i & 5i \end{pmatrix}$$

$$\Rightarrow -A = \begin{pmatrix} -5i & -3+2i \\ 3+2i & -5i \end{pmatrix} \text{-----(ii)}$$

From (i) and (ii),

$$\overline{A^T} = -A$$

Hence the above given matrix is skew-Hermitian matrix

Multiplication of Matrices:

If A and B be two matrices such that the number of columns in A is equal to the number of rows in B, that is if $A = (a_{ij})$, $B = (b_{jk})$ then the product of A and B denoted by AB.

Example 09:

$$A = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}, B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$\Rightarrow AB = \begin{pmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 4 \times 5 + 5 \times 7 & 4 \times 6 + 5 \times 8 \end{pmatrix} \text{ [Row by Column]}$$

$$\Rightarrow AB = \begin{pmatrix} 5+14 & 6+16 \\ 20+35 & 24+40 \end{pmatrix}$$

$$\therefore AB = \begin{pmatrix} 19 & 22 \\ 55 & 64 \end{pmatrix}$$

$$\begin{array}{c} \xrightarrow{\quad} \\ \xrightarrow{\quad} \end{array} A = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}, B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$\text{Example 10: } A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

Answer: Since the number of column in Matrix A is not equal to the number of rows in B. Hence AB does not exist.

$$\text{Example 11: } A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

$$\therefore AB = \begin{pmatrix} 1 \times 1 & 1 \times 2 & 1 \times 3 \\ 2 \times 1 & 2 \times 2 & 2 \times 3 \\ 3 \times 1 & 3 \times 2 & 3 \times 3 \end{pmatrix}$$

$$\therefore AB = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$$

Example 12: If $A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$ Show that $A^n = \begin{pmatrix} 1+2n & -4n \\ n & 1-2n \end{pmatrix}$; also find A^{n+1}

Answer:

$$A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{pmatrix} 3 \times 3 + (-4) \times 1 & 3 \times (-4) + (-4) \times (-1) \\ 1 \times 3 + (-1) \times 1 & 1 \times (-4) + (-1) \times (-1) \end{pmatrix}$$

$$\Rightarrow A^2 = \begin{pmatrix} 9-4 & -12+4 \\ 3-1 & -4+1 \end{pmatrix}$$

$$\Rightarrow A^2 = \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}$$

$$\Rightarrow A^2 = \begin{pmatrix} 1+2 \times 2 & -4 \times 2 \\ 2 & 1-2(2) \end{pmatrix}$$

$$\therefore A^3 = \begin{bmatrix} 1+2 \times 3 & -4 \times 3 \\ 3 & 1-2 \times 3 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$$

$$A^{n+1} = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow A^{n+1} = \begin{bmatrix} (1+2n) \times 3 + (-4n) \times 1 & (1+2n) \times (-4) + (-4n) \times (-1) \\ n \times 3 + (1-2n) \times 1 & n \times (-4) + (1-2n) \times (-1) \end{bmatrix}$$

$$\Rightarrow A^{n+1} = \begin{bmatrix} 3+6n-4n & -4-8n+4n \\ 3n+1-2n & -4n-1+2n \end{bmatrix}$$

$$\Rightarrow A^{n+1} = \begin{bmatrix} 3+2n & -4-4n \\ n+1 & -2n-1 \end{bmatrix}$$

$$\Rightarrow A^{n+1} = \begin{bmatrix} 1+2+2n & -4-4n \\ n+1 & -2n-2+1 \end{bmatrix}$$

$$\Rightarrow A^{n+1} = \begin{bmatrix} 1+2n+2 & -4n-4 \\ n+1 & 1-2n-2 \end{bmatrix}$$

$$\Rightarrow A^{n+1} = \begin{bmatrix} 1+2(n+1) & -4(n+1) \\ (n+1) & 1-2(n+1) \end{bmatrix} \text{ Answer}$$

Example 13: If $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ Show that $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$; also find A^{n+1}

Answer:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\ \Rightarrow A^2 &= \begin{pmatrix} 1 \times 1 + 1 \times 0 & 1 \times 1 + 1 \times 1 \\ 0 \times 1 + 1 \times 0 & 0 \times 1 + 1 \times 1 \end{pmatrix} \\ \Rightarrow A^2 &= \begin{pmatrix} 1+0 & 1+1 \\ 0+0 & 0+1 \end{pmatrix} \\ \Rightarrow A^2 &= \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

$$\therefore A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned} A^n &= \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \\ A^{n+1} &= \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\ \Rightarrow A^{n+1} &= \begin{bmatrix} 1 \times 1 + n \times 0 & 1 \times 1 + n \times 1 \\ 0 \times 1 + 1 \times 0 & 0 \times 0 + 1 \times 1 \end{bmatrix} \\ \Rightarrow A^{n+1} &= \begin{bmatrix} 1 & 1+n \\ 0 & 1 \end{bmatrix} \\ \Rightarrow A^{n+1} &= \begin{bmatrix} 1 & n+1 \\ 0 & 1 \end{bmatrix} \text{ Answer} \end{aligned}$$

Example 14: If ω is a cube root of unity and $A = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, then $A^{37} = ?$

Answer:

Given

$$A = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$$

$$\Rightarrow A^{37} = \begin{bmatrix} \omega^{37} & 0 \\ 0 & \omega^{37} \end{bmatrix}$$

$$\text{Now, } \omega^{37} = (\omega^3)^{12} \cdot \omega = (1)^{12} \omega = \omega$$

$$\Rightarrow A^{37} = \begin{bmatrix} \omega^{37} & 0 \\ 0 & \omega^{37} \end{bmatrix} = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix} = A$$

What is Echelon Form?

Echelon form means that the matrix is in one of two states:

- Row echelon form.
- Reduced row echelon form.

Row echelon form

01. The first number in the row (called a **leading coefficient**) is 1. *Note: some authors don't require that the leading coefficient is a 1; it could be any number.*
02. All terms below this leading entry (1 or any number) are zeros.
03. Any non-zero rows are always above rows with all zeros.
04. Rows consisting of all zeros are at the bottom of the matrix.

Example

$$\begin{array}{lll} \text{i.} & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \text{ii.} \begin{bmatrix} 1 & 3 & 6 & 2 \\ 0 & 1 & 5 & 2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \text{iii.} \begin{bmatrix} \textcircled{3} & 2 & 0 & 7 & 9 \\ 0 & \textcircled{4} & 5 & 10 & 0 \\ 0 & 0 & 0 & \textcircled{-4} & 1 \\ 0 & 0 & 0 & 0 & \textcircled{6} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

The following example is **not** in echelon form:

Matrix A does not have all-zero rows below non-zero rows.

$$\text{A.} \begin{bmatrix} 1 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

What is Reduced Row Echelon Form?

Reduced row echelon form has following requirements:

- The first non-zero number in the first row (**the leading entry**) is the number 1.
- The second row also starts with the number 1, which is further to the right than the leading entry in the first row. For every subsequent row, the number 1 must be further to the right.
- In each row, the left-most nonzero entry is 1 **and the column that contains leading entry (1) has** all other entries (below and above) equal to 0.

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 4 \\ 0 & 1 & 3 & 0 & 5 \\ 0 & 0 & 0 & 1 & 6 \end{bmatrix}$$

For each the following matrices, determine if it is in reduced row-echelon form.

i. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ ii. $\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ iii. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ iv. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ v. $\begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

i.No,

ii. No, the leading 1 in the third row is not the only nonzero in the third column containing it.

iii. No, the left most non-zero entry in the second row not equal to 1.

iv. No, the leading 1 in the second row is to the left of the leading 1 in the first row.

v. yes

Give one example of echelon form of a matrix and reduced echelon form of a matrix

$$\begin{bmatrix} 2 & 3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Reduced echelon form

What is the difference between echelon form and triangular matrix?

A triangular is a square matrix while an echelon matrix is a rectangular matrix, it is more general. Any $m \times n$ matrix can be in row-echelon form. There is a requirement for a triangular matrix to be square. That is the difference.

Example 15: A manufacturer produces three products: Toothpaste, Soap and Shampoo, which sell in the market. Annual sale volumes are indicated as follows:

Market	Products		
	Toothpaste	Soap	Shampoo
New Market	8000	10000	15000
Shopping Complex	10000	2000	20000

- If unit (mini packet) sale prices of Toothpaste, soap and shampoo are tk. 2.25, tk. 3.50 and tk. 3.00 respectively. Find the total revenue in each market with the help of matrix.
- If the unit costs of the above three products are tk. 1.60, tk. 1.90 and tk. 2.40 respectively. Find the gross profit with the help of matrices.

Solution:

The product matrix gives the total revenue in each market.

Let, Unit sale price of three products be $A = \begin{pmatrix} 2.25 & 3.50 & 3.00 \end{pmatrix}$

Annual sale of three goods in two markets be $B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$

Given,

$$A = \begin{pmatrix} 2.25 & 3.50 & 3.00 \end{pmatrix} \quad B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$$

$$\therefore AB = (2.25 \times 8000 + 3.50 \times 10000 + 3 \times 15000 \quad 2.25 \times 10000 + 3.50 \times 2000 + 3 \times 20000) \\ = (98000 \quad 89500)$$

Total revenue from the new market Tk.98, 000/-

Total revenue from Shopping Complex-Tk.89, 500/-

Total revenue from two markets: Tk. 1, 87,500/-

Similarly the costs of products with manufacturers sells in the market are:

$$A = \begin{pmatrix} 1.60 & 1.90 & 2.40 \end{pmatrix} \quad B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$$

$$\therefore AB = 1.60 \times 8000 + 1.90 \times 10000 + 2.40 \times 15000 \quad 1.60 \times 10000 + 1.90 \times 2000 + 2.40 \times 20000 \\ = (67800 \quad 67800)$$

Total cost of products for new market Tk.67, 800/-

Total cost of products for Shopping Complex-Tk.67800/-

Total cost from two markets: Tk. 1, 35,600

$$\text{So, required gross profit} = \text{Total revenue} - \text{Total Cost} = (187500 - 135600) \\ = \text{Tk.51, 900/-}$$

Ans: / Tk. 51,900/-

Example 16: A trust fund has tk. 50,000 that is to be invested into two types of bonds. The first bond pays 5% interest per year and the second bond pays 6% interest per year. Using matrix multiplication determines how to divide tk. 50,000 between the two types of bond so as to obtain an annual total interest of tk. 2780.

Solution: Let tk. 50000 be divided into two parts tk. x and tk. (50,000 – x) out of which first part is invested in first type of bonds and second part is invested in second type of bonds.

The values of these bonds can be written in the form of a row matrix $A = \begin{pmatrix} x & 50000 - x \end{pmatrix}$

The amounts received as interest per Tk. annually from these two types of bonds can be

written in the form of a column matrix $B = \begin{pmatrix} \frac{5}{100} \\ \frac{6}{100} \end{pmatrix}$

Now the interest to be obtained annually is a single number

$$\begin{aligned} AB &= (x \quad 50000 - x) \begin{pmatrix} \frac{5}{100} \\ \frac{6}{100} \end{pmatrix} \\ &= \left(x \times \frac{5}{100} + (50000 - x) \times \frac{6}{100} \right) \\ AB &= \left[\frac{x}{20} + \frac{3(50000 - x)}{50} \right] \\ &= \left[\frac{5x + 6(50000 - x)}{100} \right] \\ &= \left[\frac{5x + 300000 - 6x}{100} \right] \\ &= \left[\frac{300000 - x}{100} \right] \end{aligned}$$

Also we are given that the annual interest equal to tk. 2780. We must have,

$$\begin{aligned} [2780] &= \left[\frac{300000 - x}{100} \right] \\ \Rightarrow 2780 &= \frac{300000 - x}{100} \\ \Rightarrow 300000 - x &= 278000 \\ \Rightarrow x &= 300000 - 278000 \\ \Rightarrow x &= 22000 \end{aligned}$$

∴ one bond is taka 22000

and another bond is taka $(50000 - 22000) = \text{tk. } 28000$ Answer

Example 17: Mr. Oshubidha Ki bought 67 Pices of cake, 6 dozen apples and 7 dozen bannans for Math's Olympiad. If the price of one dozen apples is Tk. 240 the price of each dozen banana is Tk.84 and the price of a piece of cake is tk. 15. How can you calculate the total cost using matrix?

Answer:

Number of items:

Apple = 6 dozen = $6 \times 12 = 72$ pieces
 Banana = 7 dozen = $7 \times 12 = 84$ pieces
 Cake = 67 pieces

Total prices:

The price of one piece apple = $240 \div 12 = tk. 20.00$

The price of one piece banana = $84 \div 12 = tk. 7.00$

The price of a piece of cake is $tk. 15.00$

Let the cost of three products $A = [20 \ 7 \ 15]$

The amount of the product $B = \begin{bmatrix} 72 \\ 84 \\ 67 \end{bmatrix}$

$$\begin{aligned} \text{Therefore } AB &= [20 \ 7 \ 15] \times \begin{bmatrix} 72 \\ 84 \\ 67 \end{bmatrix} \\ &= [20 \times 72 + 7 \times 84 + 15 \times 67] \\ &= [1440 + 588 + 1005] \\ &= [3003] \end{aligned}$$

So total cost = Tk. 3,033

Covariance Matrix: Let the training set of M face images be $\Gamma_1, \Gamma_2, \dots, \Gamma_M$. The average face of the set is defined by:

$$\psi = \frac{1}{M} \sum_{n=1}^M \Gamma_n \text{-----(i)}$$

Each face differs from the average face by the vector $\phi_i = \Gamma_i - \psi$ where $i = 1$ to M

We shall rearrange these vectors in a matrix $A = [\phi_1 \dots \phi_M]$ of dimension $N \times M$, which will then be subject to PCA (Principal Component Analysis). The covariance Matrix is: $C = A.A^T$

Example 18:

Let image-1: $\begin{pmatrix} 4 & 6 \\ 3 & 2 \end{pmatrix}$

Let image-2: $\begin{pmatrix} 1 & 5 \\ 3 & 6 \end{pmatrix}$

Let image-3: $\begin{pmatrix} 2 & 2 \\ 4 & 9 \end{pmatrix}$

Note: MATLAB Code:

1. `i=imread('mrk.jpg')` then we will get a matrix whose entries are pixel values of the image
2. `i(:)`; then all pixel values will be arranged in a column.

Then three images make a single matrix named Γ

$$\text{Original faces (the training set), } \Gamma_i = \begin{pmatrix} 4 & 1 & 2 \\ 3 & 3 & 4 \\ 6 & 5 & 2 \\ 2 & 6 & 9 \end{pmatrix} \text{ Where } i = 1, 2, 3$$

$$\text{i.e. } \Gamma_1 = \begin{pmatrix} 4 \\ 3 \\ 6 \\ 2 \end{pmatrix}, \quad \Gamma_2 = \begin{pmatrix} 1 \\ 3 \\ 5 \\ 6 \end{pmatrix}, \quad \Gamma_3 = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 9 \end{pmatrix}$$

The average face of the set is defined by:

$$\Psi = \frac{1}{M} \sum_{N=1}^M \Gamma_n, \quad \text{Here, } M = 3$$

$$\Psi = \frac{1}{3} \sum_{N=1}^3 \Gamma_N = \frac{1}{3} (\Gamma_1 + \Gamma_2 + \Gamma_3) = \frac{1}{3} \begin{pmatrix} 4 \\ 3 \\ 6 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \\ 5 \\ 6 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 2 \\ 9 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 7 \\ 10 \\ 13 \\ 17 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix}$$

Now,

$$\phi_i = \Gamma_i - \Psi$$

$$\phi_1 = \Gamma_1 - \Psi = \begin{pmatrix} 4 \\ 3 \\ 6 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \\ -4 \end{pmatrix}$$

$$\phi_2 = \Gamma_2 - \Psi = \begin{pmatrix} 1 \\ 3 \\ 5 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\phi_3 = \Gamma_3 - \Psi = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 9 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -2 \\ 3 \end{pmatrix}$$

$$\therefore A = [\phi_1 \dots \phi_M]$$

$$A = [\phi_1 \quad \phi_2 \quad \phi_3]$$

$$\therefore A = \begin{pmatrix} 2 & -1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -2 \\ -4 & 0 & 3 \end{pmatrix}$$

$$\therefore A^T = \begin{pmatrix} 2 & 0 & 2 & -4 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 3 \end{pmatrix}$$

The covariance Matrix is:

$$C = A \cdot A^T = \begin{pmatrix} 2 & -1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -2 \\ -4 & 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 0 & 2 & -4 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 0 & 3 & -8 \\ 0 & 1 & -2 & 3 \\ 3 & -2 & 9 & -14 \\ -8 & 3 & -14 & 25 \end{pmatrix}$$

Orthogonal Matrix:

A matrix A is orthogonal if $AA^T = A^T A = I$. Where I is unit matrix and A^T is the transposed matrix of A.

Example 19: Show that the following matrix $A = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$ is orthogonal

Solution:

Given that,

$$A = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

$$\therefore A^T = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$\therefore AA^T = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$\Rightarrow AA^T = \begin{pmatrix} \cos \alpha \cdot \cos \alpha + \sin \alpha \cdot \sin \alpha & -\cos \alpha \sin \alpha + \sin \alpha \cos \alpha \\ -\sin \alpha \cos \alpha + \cos \alpha \sin \alpha & \sin \alpha \cdot \sin \alpha + \cos \alpha \cdot \cos \alpha \end{pmatrix} \quad [Row \text{ by column}]$$

$$\Rightarrow AA^T = \begin{pmatrix} \cos^2 \alpha + \sin^2 \alpha & -\cos \alpha \sin \alpha + \sin \alpha \cos \alpha \\ -\sin \alpha \cos \alpha + \cos \alpha \sin \alpha & \sin^2 \alpha + \cos^2 \alpha \end{pmatrix}$$

$$\Rightarrow AA^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \text{ (Proved)}$$

Example 20: Prove $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix}$ is orthogonal.

$$\therefore A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 \times 1 + 0 \times 0 + 0 \times 0 & 1 \times 0 + 0 \times \frac{3}{5} + 0 \times (-\frac{4}{5}) & 1 \times 0 + 0 \times \frac{4}{5} + 0 \times \frac{3}{5} \\ 0 \times 1 + \frac{3}{5} \times 0 + (-\frac{4}{5}) \times 0 & 0 \times 0 + \frac{3}{5} \times \frac{3}{5} - \frac{4}{5} \times (-\frac{4}{5}) & 0 \times 0 + \frac{3}{5} \times \frac{4}{5} - \frac{4}{5} \times \frac{3}{5} \\ 0 \times 1 + \frac{4}{5} \times 0 + \frac{3}{5} \times 0 & 0 \times 0 + \frac{4}{5} \times \frac{3}{5} + \frac{3}{5} \times (-\frac{4}{5}) & 0 \times 0 + \frac{4}{5} \times \frac{4}{5} + \frac{3}{5} \times \frac{3}{5} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 + \frac{9}{25} + \frac{16}{25} & 0 + \frac{12}{25} - \frac{12}{25} \\ 0 & 0 + \frac{12}{25} - \frac{12}{25} & 0 + \frac{16}{25} + \frac{9}{25} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 + \frac{25}{25} & 0 + 0 \\ 0 & 0 + 0 & 0 + \frac{25}{25} \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$\therefore A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \\ 0 & \frac{4}{5} & \frac{3}{5} \end{bmatrix} \text{ is orthogonal matrix.}$$

Example 21: Determine the value of α , β , γ such that the matrix $\begin{pmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & \beta & \gamma \end{pmatrix}$ is

orthogonal

$$\text{Let, } A = \begin{pmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & \beta & \gamma \end{pmatrix}$$

$$A^T = \begin{pmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & \beta \\ \gamma & -\gamma & \gamma \end{pmatrix}$$

As the given matrix is orthogonal, so $AA^T = I$

$$\begin{pmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & \beta & \gamma \end{pmatrix} \begin{pmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & \beta \\ \gamma & -\gamma & \gamma \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0+4\beta^2+\gamma^2 & 0+2\beta^2-\gamma^2 & 0+2\beta^2+\gamma^2 \\ 0+2\beta^2-\gamma^2 & \alpha^2+\beta^2+\gamma^2 & \alpha^2+\beta^2-\gamma^2 \\ 0+2\beta^2+\gamma^2 & \alpha^2+\beta^2-\gamma^2 & \alpha^2+\beta^2+\gamma^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Thus we can obtain the following system of equations

$$4\beta^2 + \gamma^2 = 1 \quad \text{-----(i)}$$

$$2\beta^2 - \gamma^2 = 0 \quad \text{-----(ii)}$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 \quad \text{-----(iii)}$$

From (ii),

$$2\beta^2 - \gamma^2 = 0$$

$$\text{Or, } 2\beta^2 = \gamma^2$$

From (i),

$$4\beta^2 + \gamma^2 = 1$$

$$\text{Or, } 2(2\beta^2) + \gamma^2 = 1$$

$$\text{Or, } 2\gamma^2 + \gamma^2 = 1$$

$$\text{Or, } 3\gamma^2 = 1$$

$$\text{Or, } \gamma^2 = \frac{1}{3}$$

$$\text{Or, } \gamma = \frac{1}{\sqrt{3}}$$

Again, from (ii),

$$2\beta^2 - \gamma^2 = 0$$

$$\text{Or, } 2\beta^2 = \gamma^2$$

$$\text{Or, } \beta^2 = \frac{1}{2} \gamma^2$$

$$\text{Or, } \beta^2 = \frac{1}{2} \cdot \frac{1}{3}$$

$$\text{Or, } \beta^2 = \frac{1}{6}$$

$$\text{Or, } \beta = \frac{1}{\sqrt{6}}$$

Again, from (iii),

$$\alpha^2 + \beta^2 + \gamma^2 = 1$$

$$\text{Or, } \alpha^2 = 1 - \beta^2 - \gamma^2$$

$$\text{Or, } \alpha^2 = 1 - \frac{1}{6} - \frac{1}{3}$$

$$\text{Or, } \alpha^2 = \frac{6-1-2}{6}$$

$$\text{Or, } \alpha^2 = \frac{3}{6}$$

$$\text{Or, } \alpha^2 = \frac{1}{2}$$

$$\text{Or, } \alpha = \frac{1}{\sqrt{2}}$$

So the values of α, β and γ are $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{3}}$ Answer

Adjoint of a Matrix:

$$\mathbf{A} = (\mathbf{a}_{ij}) = \begin{pmatrix} \mathbf{a}_{11} & \mathbf{a}_{12} & \mathbf{a}_{13} & - & - & - & - & - & \mathbf{a}_{1n} \\ \mathbf{a}_{21} & \mathbf{a}_{22} & \mathbf{a}_{23} & - & - & - & - & - & \mathbf{a}_{2n} \\ \mathbf{a}_{31} & \mathbf{a}_{32} & \mathbf{a}_{33} & - & - & - & - & - & \mathbf{a}_{3n} \\ - & - & - & - & - & - & - & - & - \\ \mathbf{a}_{m1} & \mathbf{a}_{m2} & \mathbf{a}_{m3} & - & - & - & - & - & \mathbf{a}_{mn} \end{pmatrix}$$

Let $\mathbf{c}_{ij}$ be the cofactor of the element $\mathbf{a}_{ij}$ in $|\mathbf{A}| = |\mathbf{a}_{ij}|$ of the $\mathbf{m} \times \mathbf{n}$ matrix $\mathbf{A} = (\mathbf{a}_{ij})$. Then

$$\mathbf{A}^c = (\mathbf{c}_{ij}) = \begin{pmatrix} \mathbf{c}_{11} & \mathbf{c}_{12} & \mathbf{c}_{13} & - & - & - & - & - & \mathbf{c}_{1n} \\ \mathbf{c}_{21} & \mathbf{c}_{22} & \mathbf{c}_{23} & - & - & - & - & - & \mathbf{c}_{2n} \\ \mathbf{c}_{31} & \mathbf{c}_{32} & \mathbf{c}_{33} & - & - & - & - & - & \mathbf{c}_{3n} \\ - & - & - & - & - & - & - & - & - \\ \mathbf{c}_{m1} & \mathbf{c}_{m2} & \mathbf{c}_{m3} & - & - & - & - & - & \mathbf{c}_{mn} \end{pmatrix}$$

[$\mathbf{c}_{11}$ =cofactor of $\mathbf{a}_{11}$, $\mathbf{c}_{12}$ =cofactor of $\mathbf{a}_{12}$].

$$\text{Adjoint of } A = (A^c)^t = \begin{pmatrix} c_{11} & c_{21} & c_{31} & - & - & - & - & - & c_{m1} \\ c_{12} & c_{22} & c_{32} & - & - & - & - & - & c_{m2} \\ c_{13} & c_{23} & c_{33} & - & - & - & - & - & c_{m3} \\ - & - & - & - & - & - & - & - & - \\ c_{1n} & c_{2n} & c_{3n} & - & - & - & - & - & c_{mn} \end{pmatrix}$$

Let,

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \text{----- (i) [Where } i = 1,2,3 \text{ \& } j = 1,2,3]$$

$$A = (a_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$[(-1)^{i+j}] = (-1)^{1+1} = (-1)^2 = +1]$$

$$[(-1)^{i+j}] = (-1)^{1+2} = (-1)^3 = -1]$$

$$\therefore \text{Co-factor of } a_{11} = c_{11} = + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{12} = c_{12} = - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{13} = c_{13} = + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{21} = c_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{22} = c_{22} = + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{23} = c_{23} = - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{31} = c_{31} = + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{32} = c_{32} = - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{33} = c_{33} = + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$\begin{array}{c} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \\ \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} \end{array}$$

$$\text{Co-factor Matrix } A^C = (c_{ij}) = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

$$\text{Transpose of a Matrix: } (A^C)^T = (c_{ji}) = \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix}$$

$$\text{Hence Adjoint Matrix of A} = \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix}$$

$$\text{Example 22: Find the adjoint of a matrix } A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{bmatrix}$$

We have,

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \text{Where } i = 1,2,3 \text{ and Where } j = 1,2,3$$

$$A = (a_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$\therefore \text{Co-factor of } a_{11} = c_{11} = + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{12} = c_{12} = - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{13} = c_{13} = + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{21} = c_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{22} = c_{22} = + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{23} = c_{23} = - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{31} = c_{31} = + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{32} = c_{32} = - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$$

$$\therefore \text{Co-factor of } a_{33} = c_{33} = + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$\text{Here, given, } A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & -6 & -7 \end{bmatrix} \text{-----(i)}$$

$$\text{We know } A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \text{-----(ii)}$$

Comparing (i) and (ii)

Here, $a_{11} = 1$, $a_{12} = 0$, $a_{13} = -1$, $a_{21} = 3$, $a_{22} = 4$, $a_{23} = 5$, $a_{31} = 0$, $a_{32} = -6$, $a_{33} = -7$

Now,

$$c_{11} = + \begin{vmatrix} 4 & 5 \\ -6 & -7 \end{vmatrix} = 2 \quad c_{12} = - \begin{vmatrix} 3 & 5 \\ 0 & -7 \end{vmatrix} = 21 \quad c_{13} = + \begin{vmatrix} 3 & 4 \\ 0 & -6 \end{vmatrix} = -18$$

$$c_{21} = - \begin{vmatrix} 0 & -1 \\ -6 & -7 \end{vmatrix} = 6 \quad c_{22} = + \begin{vmatrix} 1 & -1 \\ 0 & -7 \end{vmatrix} = -7 \quad c_{23} = - \begin{vmatrix} 1 & 0 \\ 0 & -6 \end{vmatrix} = 6$$

$$c_{31} = + \begin{vmatrix} 0 & -1 \\ 4 & 5 \end{vmatrix} = 4 \quad c_{32} = - \begin{vmatrix} 1 & -1 \\ 3 & 5 \end{vmatrix} = -8 \quad c_{33} = + \begin{vmatrix} 1 & 0 \\ 3 & 4 \end{vmatrix} = 4$$

$$\text{Co-factor Matrix } A^C = (c_{ij}) = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} = \begin{bmatrix} 2 & 21 & -18 \\ 6 & -7 & 6 \\ 4 & -8 & 4 \end{bmatrix}$$

$$\therefore \text{Adj. } A = (A^C)^T = \begin{bmatrix} 2 & 6 & 4 \\ 21 & -7 & -8 \\ -18 & 6 & 4 \end{bmatrix}$$

Inverse Matrix:

The inverse of a matrix, $A^{-1} = \frac{\text{Adjoint of } A}{|A|}$

$|A| \rightarrow$ The determinant of the matrix A.

Example 23: Find the Inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ where $|A| \neq 0$

Solution:

$$|A| = 1(4 \times 6 - 5 \times 5) - 2(2 \times 6 - 3 \times 5) + 3(2 \times 5 - 3 \times 4)$$

$$|A| = 1(24 - 25) - 2(12 - 15) + 3(10 - 12)$$

$$= 1(-1) - 2(-3) + 3(-2)$$

$$= -1 + 6 - 6$$

$$= -1$$

We have

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

$$A = (a_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Now,

$$c_{11} = + \begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} = -1 \quad c_{12} = - \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix} = 3 \quad c_{13} = + \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = -2$$

$$c_{21} = - \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = 3 \quad c_{22} = + \begin{vmatrix} 1 & 3 \\ 3 & 6 \end{vmatrix} = -3 \quad c_{23} = - \begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = 1$$

$$c_{31} = + \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -2 \quad c_{32} = - \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = 1 \quad c_{33} = + \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$$

$$A^C = \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix}$$

$$\text{Adjoint of } A = (A^C)^T = \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix}$$

$$A^{-1} = \frac{\text{Adjoint of } A}{|A|} = \frac{1}{-1} \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix} = -1 \begin{pmatrix} -1 & 3 & -2 \\ 3 & -3 & 1 \\ -2 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{pmatrix}$$

Perfection Test:

$$AA^{-1} = I$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} ; A^{-1} = \begin{pmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{pmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1 \times 1 + 2 \times (-3) + 3 \times 2 & 1 \times (-3) + 2 \times 3 + 3 \times (-1) & 1 \times 2 + 2 \times (-1) + 3 \times 0 \\ 2 \times 1 + 4 \times (-3) + 5 \times 2 & 2 \times (-3) + 4 \times 3 + 5 \times (-1) & 2 \times 2 + 4 \times (-1) + 5 \times 0 \\ 3 \times 1 + 5 \times (-3) + 6 \times 2 & 3 \times (-3) + 5 \times 3 + 6 \times (-1) & 3 \times 2 + 5 \times (-1) + 6 \times 0 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1-6+6 & -3+6-3 & 2-2+0 \\ 2-12+10 & -6+12-5 & 4-4+0 \\ 3-15+12 & -9+15-6 & 6-5+0 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Example 24: Find the inverse Matrix of

$$A = \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix} \text{ Where } [a^2 + b^2 + c^2 + d^2 = 1]$$

Here, $a_{11} = a+ib$, $a_{12} = c+id$, $a_{21} = -c+id$, $a_{22} = a-ib$

$$\therefore c_{11} = \text{cofactor of } a+ib = +(a-ib)$$

$$c_{12} = \text{cofactor of } c+id = -(-c+id)$$

$$c_{21} = \text{cofactor of } -c+id = -(c+id)$$

$$c_{22} = \text{cofactor of } a-ib = +(a+ib)$$

So, $c_{11} = a-ib$, $c_{12} = c-id$, $c_{21} = -c-id$, $c_{22} = a+ib$

$$A^C = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} a-ib & c-id \\ -c-id & a+ib \end{pmatrix}$$

$$\text{Adj}A = (A^C)^T = \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix}$$

$$\text{Given } A = \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix}$$

$$\begin{aligned} \therefore |A| &= \begin{vmatrix} a+ib & c+id \\ -c+id & a-ib \end{vmatrix} = (a+ib)(a-ib) - (c+id)(-c+id) \\ &= (a^2 - iab + iab - i^2b^2) - (-c^2 + icd - icd + i^2d^2) \\ &= (a^2 - iab + iab + b^2) - (-c^2 + icd - icd - d^2) [\because i^2 = -1] \\ &= (a^2 + b^2 + c^2 + d^2) \\ &= 1 [\because a^2 + b^2 + c^2 + d^2 = 1] \end{aligned}$$

$$\therefore A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{1}{1} \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix} = \begin{pmatrix} a-ib & -c-id \\ c-id & a+ib \end{pmatrix}$$

Perfection Test:

We have,

$$A = \begin{pmatrix} a + ib & c + id \\ -c + id & a - ib \end{pmatrix} \text{ and } A^{-1} = \begin{pmatrix} a - ib & -c - id \\ c - id & a + ib \end{pmatrix}$$

$$AA^{-1} = \begin{pmatrix} a + ib & c + id \\ -c + id & a - ib \end{pmatrix} \begin{pmatrix} a - ib & -c - id \\ c - id & a + ib \end{pmatrix}$$

$$= \begin{pmatrix} (a + ib)(a - ib) + (c + id)(c - id) & (a + ib)(-c - id) + (c + id)(a + ib) \\ (-c + id)(a - ib) + (a - ib)(c - id) & (-c + id)(-c - id) + (a - ib)(a + ib) \end{pmatrix}$$

$$= \begin{pmatrix} a^2 - (ib)^2 + c^2 - (id)^2 & -ac - iad - ibc + bd + ac + ibc + iad - bd \\ -ac + ibc + iad + bd + ac - iad - ibc - bd & (-c)^2 - (id)^2 + (a)^2 - (ib)^2 \end{pmatrix}$$

$$[\because (a + b)(a - b) = a^2 - b^2]$$

$$= \begin{pmatrix} a^2 + b^2 + c^2 + d^2 & 0 \\ 0 & a^2 + b^2 + c^2 + d^2 \end{pmatrix} \quad [i^2 = -1]$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad [a^2 + b^2 + c^2 + d^2 = 1]$$

Characteristics Matrix of 'A':

The Matrix $A - \lambda I$ is called the Characteristics matrix of 'A' where I is the Identity matrix

$$\text{Let, } A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad [I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}]$$

$$A - \lambda I = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 2 \\ 3 & 2 - \lambda \end{pmatrix} ; \text{ Which is called Characteristics Matrix of A}$$

Characteristics Polynomial of 'A':

The determinant $|A - \lambda I|$ is called the Characteristics Polynomial of A

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 2 \\ 3 & 2 - \lambda \end{vmatrix}$$

$$= (1 - \lambda)(2 - \lambda) - 6 = 2 - \lambda - 2\lambda + \lambda^2 - 6$$

$$= \lambda^2 - 3\lambda - 4 ; \text{ which is polynomial of degree 2}$$

Characteristics Equation of 'A':

The equation $|A - \lambda I| = 0$ is known as the Characteristics equation of 'A'

$$\begin{aligned}
 |A - \lambda I| &= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0 \\
 &= (1-\lambda)(2-\lambda) - 6 = 0 = 2 - \lambda - 2\lambda + \lambda^2 - 6 = 0 \\
 &= \lambda^2 - 3\lambda - 4 = 0; \text{ which is Characteristic equation of the matrix } A
 \end{aligned}$$

Characteristics root or latent roots or Eigen values:

Its roots are called Characteristics root or latent roots or Eigen values of Characteristics values.

$$\begin{aligned}
 |A - \lambda I| &= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0 \\
 &= (1-\lambda)(2-\lambda) - 6 = 0 = 2 - \lambda - 2\lambda + \lambda^2 - 6 = 0 \\
 &= \lambda^2 - 3\lambda - 4 = 0
 \end{aligned}$$

$\lambda = 4, -1$; These are Characteristics root or latent roots or Eigen values of A.

Spectrum of A:

The set of all the Eigen values of matrix is called the spectrum of 'A'.

$\{4, -1\}$ is the spectrum of A.

The set of eigenvalues is called the spectrum of A in analogy with the idea of the spectrum of light and the composed by various wavelengths with different intensity. For this reason, the procedure is often referred to as a spectral decomposition. If two or more eigenvalues of A are identical, the spectrum of the matrix is called degenerate.

Example 25: Find the Characteristics root or latent roots or Eigen values of a matrix

$$\begin{aligned}
 A &= \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix} \\
 A - \lambda I &= \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
 A - \lambda I &= \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\
 A - \lambda I &= \begin{pmatrix} \cos \theta - \lambda & -\sin \theta \\ -\sin \theta & -\cos \theta - \lambda \end{pmatrix} \\
 \therefore |A - \lambda I| &= \begin{vmatrix} \cos \theta - \lambda & -\sin \theta \\ -\sin \theta & -\cos \theta - \lambda \end{vmatrix} \\
 &= (\cos \theta - \lambda)(-\cos \theta - \lambda) - (-\sin \theta)(-\sin \theta) \\
 &= -\cos^2 \theta - \lambda \cos \theta + \lambda \cos \theta + \lambda^2 - \sin^2 \theta \\
 &= \lambda^2 - (\cos^2 \theta + \sin^2 \theta) \\
 &= \lambda^2 - 1 [\because \cos^2 \theta + \sin^2 \theta = 1]
 \end{aligned}$$

$$\therefore |A - \lambda I| = \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 = 1$$

$$\Rightarrow \lambda = \pm 1$$

Linear System of Equations:

Let a system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ has the general form:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = b_3$$

$$\dots$$

$$\dots$$

$$\dots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

Then this system can be written in the matrix form as $AX = b$

$$\text{Where, } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & - & - & - & - & - & a_{1n} \\ a_{21} & a_{22} & a_{23} & - & - & - & - & - & a_{2n} \\ a_{31} & a_{32} & a_{33} & - & - & - & - & - & a_{3n} \\ - & - & - & - & - & - & - & - & - \\ a_{m1} & a_{m2} & a_{m3} & - & - & - & - & - & a_{mn} \end{pmatrix}; \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ - \\ x_n \end{pmatrix}; \quad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ - \\ b_n \end{pmatrix}$$

We have, $AX = b$

Multiplying by A^{-1} on both sides

$$AA^{-1}X = A^{-1}b$$

$$\Rightarrow IX = A^{-1}b \quad [\because AA^{-1} = I]$$

$$\Rightarrow X = A^{-1}b \quad [\because IX = X]$$

Example:

$$\text{Let, } I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned}
\therefore \mathbf{IX} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \times x + 0 \times y + 0 \times z \\ 0 \times x + 1 \times y + 0 \times z \\ 0 \times x + 0 \times y + 1 \times z \end{pmatrix} \\
&= \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\
&= \mathbf{X} \\
\therefore \mathbf{IX} &= \mathbf{X}
\end{aligned}$$

Example 26: Solve by Matrix method:

$$x + 2y + 3z = 14$$

$$3x + y + 2z = 11$$

$$2x + 3y + z = 11$$

Let the system can be written $\mathbf{AX} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}; \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \quad \mathbf{b} = \begin{bmatrix} 14 \\ 11 \\ 11 \end{bmatrix}$$

We have, $\mathbf{AX} = \mathbf{b}$

Multiplying by $\mathbf{A}^{-1}$ on both sides

$$\mathbf{AA}^{-1}\mathbf{X} = \mathbf{A}^{-1}\mathbf{b}$$

$$\Rightarrow \mathbf{IX} = \mathbf{A}^{-1}\mathbf{b} \quad [\because \mathbf{AA}^{-1} = \mathbf{I}]$$

$$\Rightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{b} \quad [\because \mathbf{IX} = \mathbf{X}]$$

$$\Rightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{b}$$

$$\Rightarrow \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{A}^{-1}\mathbf{b} \quad \text{------(i)} \quad [\because \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}]$$

Also we have, $\mathbf{A}^{-1} = \frac{\mathbf{AdjA}}{|\mathbf{A}|}$

$$\text{We have, } \mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$$

$$\mathbf{A} = (\mathbf{a}_{ij}) = \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Now,

$$\mathbf{c}_{11} = + \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -5; \quad \mathbf{c}_{12} = - \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = 1; \quad \mathbf{c}_{13} = + \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} = 7;$$

$$\mathbf{c}_{21} = - \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix} = 7; \quad \mathbf{c}_{22} = + \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = -5; \quad \mathbf{c}_{23} = - \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 1;$$

$$c_{31} = + \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1; \quad c_{32} = - \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} = 7; \quad c_{33} = + \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -5;$$

The co-factor Matrix of A: $A^C = \begin{bmatrix} -5 & 1 & 7 \\ 7 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}$

$$\text{Adj. A} = (A^C)^T = \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix}$$

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{vmatrix} = 1(1 \times 1 - 2 \times 3) - 2(3 \times 1 - 2 \times 2) + 3(3 \times 3 - 2 \times 1) \\ &= 1 \times (-5) - 2 \times (-1) + 3 \times 7 \\ &= -5 + 2 + 21 \\ &= 18 \end{aligned}$$

$$A^{-1} = \frac{\text{Adj}A}{|A|} = \frac{1}{18} \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix}$$

From (i),

$$\Rightarrow X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}b$$

$$\therefore X = A^{-1}b = \frac{1}{18} \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix} \begin{bmatrix} 14 \\ 11 \\ 11 \end{bmatrix}$$

$$X = \frac{1}{18} \begin{bmatrix} -5 \times 14 + 7 \times 11 + 1 \times 11 \\ 1 \times 14 - 5 \times 11 + 7 \times 11 \\ 7 \times 14 + 1 \times 11 - 5 \times 11 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{18} \begin{bmatrix} -70 + 77 + 11 \\ 14 - 55 + 77 \\ 98 + 11 - 55 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 18 \\ 36 \\ 54 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 2, z = 3 \text{ Answer}$$

Example 27:

Find x, y, z, t using the concept of equality of matrices, where

$$3 \begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2t \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+t & 3 \end{bmatrix}$$

Rank of a Matrix:

The number of non zero rows of a matrix given in the echolon form is its rank.

The number of linear independent rows of the matrix is called its rank.

Example

$$2x + 3y + z = 1$$

$$4x + 5y + 7z = 9$$

$$x + 9y + z = 0$$

$$4x + 6y + 2z = 2 \text{ -----Dependent}$$

$$\text{Let, } A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 5 & 7 \\ 1 & 9 & 1 \\ 4 & 6 & 2 \end{pmatrix}$$

$$[r'_4 = r_4 - 2r_1]$$

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 5 & 7 \\ 1 & 9 & 1 \\ 4 & 6 & 2 \end{pmatrix} \approx \begin{pmatrix} 2 & 3 & 1 \\ 4 & 5 & 7 \\ 1 & 9 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The number of non zero rows of the matrix is 3.

Hence, it's rank is 3.

Example 28: Find the rank of the matrix $A = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 4 & 6 \\ 0 & 6 & 9 \end{bmatrix}$

$$A \sim \begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [r'_2 = r_2 - 2r_1]; [r'_3 = r_3 - 3r_1]$$

Then the rank of matrix is one. $[\because 1 \text{ non zero row}]$

Example 29: Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 1 & 2 \end{bmatrix}$

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 4 & 1 & 6 \\ 2 & 1 & 2 \end{bmatrix} \quad [C'_2 = C_3 - C_2]$$

$$A \sim \begin{bmatrix} 3 & 0 & 3 \\ 4 & 1 & 6 \\ 2 & 0 & 4 \end{bmatrix} \quad [r'_1 = r_2 - r_1]; [r'_3 = r_2 - r_3]$$

$$A \sim \begin{bmatrix} 0 & 0 & 3 \\ 2 & 1 & 6 \\ 2 & 0 & 4 \end{bmatrix} \quad [C'_1 = C_3 - C_1]$$

$$A \sim \begin{bmatrix} 0 & 0 & 3 \\ 0 & 1 & 6 \\ -2 & 0 & 4 \end{bmatrix} \quad [C'_1 = C_3 - 3C_1]$$

$$A \sim \begin{bmatrix} -2 & 0 & 4 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

Then the rank of matrix is 3.

Example 30: Let, $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ Given, $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

Prove that, $(cI + bE)^3 = c^3I + 3c^2bE$

L.H.S.

$$cI = c \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix}$$

$$bE = b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$$

$$\therefore cI + bE = \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix} + \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$$

$$\begin{aligned}
\Rightarrow cI + bE &= \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^2 &= \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^2 &= \begin{pmatrix} c \times c + b \times 0 & c \times b + b \times c \\ 0 \times c + c \times 0 & 0 \times b + c \times c \end{pmatrix} \\
\Rightarrow (cI + bE)^2 &= \begin{pmatrix} c^2 & 2bc \\ 0 & c^2 \end{pmatrix} \\
\Rightarrow (cI + bE)^2 (cI + bE) &= \begin{pmatrix} c^2 & 2bc \\ 0 & c^2 \end{pmatrix} \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^3 &= \begin{pmatrix} c^2 & 2bc \\ 0 & c^2 \end{pmatrix} \begin{pmatrix} c & b \\ 0 & c \end{pmatrix} \\
\Rightarrow (cI + bE)^3 &= \begin{pmatrix} c^2 \times c + 2bc \times 0 & c^2 \times b + 2bc \times c \\ 0 \times c + c^2 \times 0 & 0 \times b + c^2 \times c \end{pmatrix} \\
\Rightarrow (cI + bE)^3 &= \begin{pmatrix} c^3 & 3bc^2 \\ 0 & c^3 \end{pmatrix} \\
\text{R.H.S. } c^3I + 3c^2bE &= c^3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 3c^2b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} c^3 & 0 \\ 0 & c^3 \end{pmatrix} + \begin{pmatrix} 0 & 3c^2b \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} c^3 & 3c^2b \\ 0 & c^3 \end{pmatrix} \text{ (Proved)}
\end{aligned}$$

Linear Equation:

By a linear equation over the real field 'R' we meant an expression $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b$ where $a_i, b_i \in \mathbf{R}$ and x_i are in determinants (or unknown or variables).

Example: $2x_1 + 3x_2 + 4x_3 = 3$

System of linear equations:

We now consider a system of m linear equations in the n unknown $x_1, x_2, x_3, \dots, x_n$

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n &= b_2 \\
a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n &= b_3 \\
\dots &\dots \\
\dots &\dots \\
\dots &\dots \\
a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n &= b_m
\end{aligned}$$

Where $a_i, b_i \in \mathbb{R}$.

The system is said to be homogeneous. If the constants $b_1, b_2, b_3, \dots, b_m$ are all zero. Otherwise it is called non-homogeneous system of equations

Example:
$$\left. \begin{aligned} 2x + 3y + z &= 7 \\ 4x + 5y + 7z &= 9 \\ x + 9y + z &= 0 \end{aligned} \right\} \longrightarrow \text{Non-homogeneous system of Linear equations}$$

Solution of a System of Linear Equations:

Step 1: Interchange equations so that the first unknown x_1 has a nonzero coefficient in the first equation, that is, so that $a_{11} \neq 0$

Step 2: To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

Step 3: If an equation $0 \times x_1 + 0 \times x_2 + 0 \times x_3 + \dots + 0 \times x_n = b, b \neq 0$ occurs the System is inconsistent and has no solution.

Step 4: If an equation $0 \times x_1 + 0 \times x_2 + 0 \times x_3 + \dots + 0 \times x_n = 0$ occurs, then the Equation can be deleted without affecting the solution.

Step 5: Solution of non-homogeneous system of equation.

- i. If $r = n$, that is there are as many equations as unknowns then the system has a unique solution.
- ii. If $r < n$, that is there are fewer equations than unknowns then we can arbitrarily assign values to the $n - r$ free variables and obtain more than one solution. of the system.

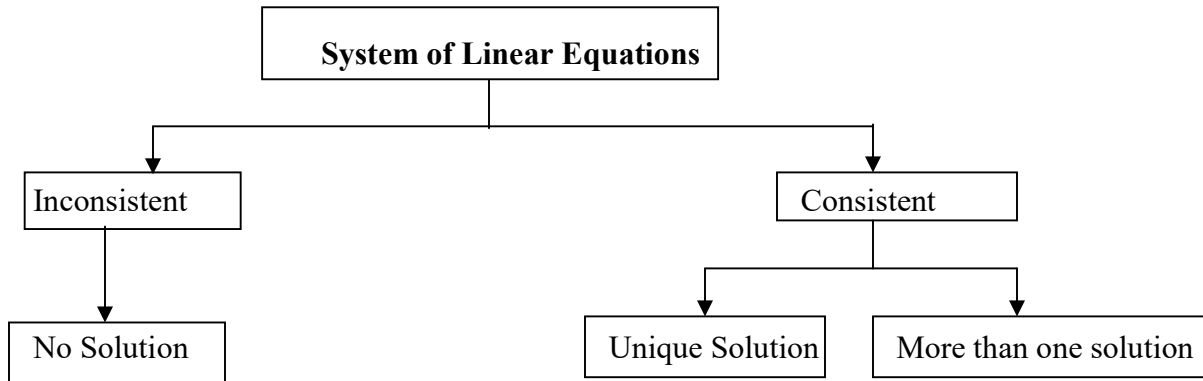
Step 6: A solution of a homogeneous system of linear equations.

- i. If $r = n$, then the system has only the zero solution.
- ii. If $r < n$, then the system has more than one non-zero solution.

$n \rightarrow$ Number of variables

$r \rightarrow$ Number of Rows or equation

Step 7: If two equations or more are identical. We can ignore them keeping one equation



Example 31: Solve the system,

$$2x + y - 2z = 10 \text{ -----}L_1$$

$$3x + 2y + 2z = 1 \text{ -----}L_2$$

$$5x + 4y + 3z = 4 \text{ -----}L_3$$

Here,

$$\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ \uparrow & \uparrow & \uparrow \\ 2x + 1y - 2z = 10 \end{array}$$

$$\begin{array}{ccc} a_{21} & a_{22} & a_{23} \\ \uparrow & \uparrow & \uparrow \\ 3x + 2y + 2z = 1 \end{array}$$

$$\begin{array}{ccc} a_{31} & a_{32} & a_{33} \\ \uparrow & \uparrow & \uparrow \\ 5x + 4y + 3z = 4 \end{array}$$

Here, $a_{11} = 2$, $a_{12} = 1$, $a_{13} = -2$, $a_{21} = 3$, $a_{22} = 2$, $a_{23} = 2$, $a_{31} = 5$, $a_{32} = 4$, $a_{33} = 3$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

1st time:

$$\begin{aligned} i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\ &= -3(2x + y - 2z - 10) + 2(3x + 2y + 2z - 1) \\ &= -6x - 3y + 6z + 30 + 6x + 4y + 4z - 2 \\ &= y + 10z + 28 \end{aligned}$$

$$\begin{aligned} i = 3, L_3 &\rightarrow -a_{31}L_1 + a_{11}L_3 \\ &= -5(2x + y - 2z - 10) + 2(5x + 4y + 3z - 4) \\ &= -10x - 5y + 10z + 50 + 10x + 8y + 6z - 8 \\ &= 3y + 16z + 42 \end{aligned}$$

Thus we obtain the following new system

$$2x + y - 2z = 10$$

$$y + 10z = -28$$

$$3y + 16z = -42$$

2nd time:

$$2x + y - 2z = 10$$

$$y + 10z = -28 \text{ -----} L_1$$

$$3y + 16z = -42 \text{ -----} L_2$$

Here, $a_{11} = 1$, $a_{12} = 10$, $a_{21} = 3$, $a_{22} = 16$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -3(y + 10z + 28) + 1(3y + 16z + 42)$$

$$= -3y - 30z - 84 + 3y + 16z + 42$$

$$= -14z - 42$$

Thus we obtain the following system

$$2x + y - 2z = 10$$

$$y + 10z = -28$$

$$-14z = 42$$

Now $-14z = 42$

$$\therefore z = -3$$

Again,

$$y + 10z = -28$$

$$\Rightarrow y + 10(-3) = -28$$

$$\Rightarrow y = -28 + 30$$

$$\therefore y = 2$$

Again,

$$2x + y - 2z = 10$$

$$\Rightarrow 2x + 2 - 2(-3) = 10$$

$$\Rightarrow 2x + 2 + 6 = 10$$

$$\Rightarrow 2x = 2$$

$$\therefore x = 1$$

Ans: / (x , y , z) = (1 , 2 , -3)

$\begin{array}{cc} \uparrow a_{11} & \uparrow a_{12} \\ 1y + 10z = -28 \\ \uparrow a_{21} & \uparrow a_{22} \\ 3y + 16z = -42 \end{array}$
--

Example 32: Solve the system,

$$2x + y - 3z = 5 \text{ -----} L_1$$

$$3x - 2y + 2z = 5 \text{ -----} L_2$$

$$5x - 3y - z = 16 \text{ -----} L_3$$

Sol:

1st time:

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{aligned}
&= -3(2x + y - 3z - 5) + 2(3x - 2y + 2z - 5) \\
&= -6x - 3y + 9z + 15 + 6x - 4y + 4z - 10 \\
&= -7y + 13z + 5
\end{aligned}$$

$$\begin{aligned}
i = 3, L_3 &\rightarrow -a_{31}L_1 + a_{11}L_3 \\
&= -5(2x + y - 3z - 5) + 2(5x - 3y - z - 16) \\
&= -10x - 5y + 15z + 25 + 10x - 6y - 2z - 32 \\
&= -11y + 13z - 7
\end{aligned}$$

Thus we obtain the following new system

$$\begin{aligned}
2x + y - 3z &= 5 \\
-7y + 13z &= -5 \\
-11y + 13z &= 7
\end{aligned}$$

2nd time:

$$\begin{aligned}
2x + y - 3z &= 5 \\
-7y + 13z &= -5 \text{ ----- } L_1 \\
-11y + 13z &= 7 \text{ ----- } L_2
\end{aligned}$$

Here, $a_{11} = -7$, $a_{12} = 13$, $a_{21} = -11$, $a_{22} = 13$

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$\begin{aligned}
i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\
&= -(-11)(-7y + 13z + 5) + (-7)(-11y + 13z - 7) \\
&= -77y + 143z + 55 + 77y - 91z + 49 \\
&= 52z + 104
\end{aligned}$$

Thus we obtain the following system

$$\begin{aligned}
2x + y - 3z &= 5 \\
-7y + 13z &= -5 \\
52z &= -104
\end{aligned}$$

Now,

$$52z = -104$$

$$\Rightarrow z = -\frac{104}{52}$$

$$\therefore z = -2$$

And

$$\begin{aligned}
-7y + 13z &= -5 \\
-7y - 13 \times 2 &= -5 \\
\Rightarrow -7y &= -5 + 26 \\
\Rightarrow y &= \frac{21}{-7} \\
\therefore y &= -3
\end{aligned}$$

$ \begin{array}{cc} \begin{array}{c} a_{11} \\ \uparrow \end{array} & \begin{array}{c} a_{12} \\ \uparrow \end{array} \\ -7y + 13z = -5 \\ \begin{array}{cc} \begin{array}{c} a_{21} \\ \uparrow \end{array} & \begin{array}{c} a_{22} \\ \uparrow \end{array} \\ -11y + 13z = 7 \end{array} \end{array} $
--

And, $2x + y - 3z = 5$

$$2x - 3 + 6 = 5$$

$$\Rightarrow 2x = 5 - 3$$

$$\Rightarrow 2x = 2$$

$$\therefore x = 1$$

Ans: / $(x, y, z) = (1, -3, -2)$

Example 33: Solve the system:

$$2x + 3y - 2z = 5 \text{ -----} L_1$$

$$x - 2y + 3z = 2 \text{ -----} L_2$$

$$4x - y + 4z = 1 \text{ -----} L_3$$

Sol: 1st time:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -1(2x + 3y - 2z - 5) + 2(x - 2y + 3z - 2)$$

$$= -2x - 3y + 2z + 5 + 2x - 4y + 6z - 4$$

$$= -7y + 8z + 1$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -4(2x + 3y - 2z - 5) + 2(4x - y + 4z - 1)$$

$$= -8x - 12y + 8z + 20 + 8x - 2y + 8z - 2$$

$$= -14y + 16z + 18$$

Thus we obtain the following system

$$2x + 3y - 2z = 5$$

$$-7y + 8z = -1$$

$$-14y + 16z = -18$$

Again, **2nd time:**

$$2x + 3y - 2z = 5$$

$$-7y + 8z = -1 \text{ -----} L_1$$

$$-14y + 16z = -18 \text{ -----} L_2$$

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -(-14)(-7y + 8z + 1) + (-7)(-14y + 16z + 18)$$

$$= -98y + 112z + 14 + 98y - 112z - 126$$

$$= -112$$

Thus we obtain the following system

$$2x + 3y - 2z = 5$$

$$-7y + 8z = -1$$

$$0 = 112$$

Here,

$ \begin{array}{cc} a_{11} & a_{12} \\ \uparrow & \uparrow \\ -7y + 8z = -1 \end{array} $
$ \begin{array}{cc} a_{21} & a_{22} \\ \uparrow & \uparrow \\ -14y + 16z = -18 \end{array} $

$$0 \times x + 0 \times y + 0 \times z = 112$$

$$\Rightarrow 0 = 112$$

$\Rightarrow$ but $112 \neq 0$; It is impossible (inconsistent)

So, the system is inconsistent and has no solution (*see step-03*)

Ans: The system has no solution

How to block diagonal of a matrix

Let a matrix

$$\begin{bmatrix} 3 & 7 & 4 \\ 2 & 5 & 1 \\ 8 & 6 & 9 \end{bmatrix}$$

Multiplying this matrix by a column matrix: $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\begin{bmatrix} 3 & 7 & 4 \\ 2 & 5 & 1 \\ 8 & 6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x + 7y + 4z \\ 2x + 5y + z \\ 8x + 6y + 9z \end{bmatrix}$$

Example 34: Solve the system,

$$x + 2y - 3z = 6$$

$$2x - y + 4z = 2$$

$$4x + 3y - 2z = 14$$

Sol: We have,

1st time:

$$L_i \rightarrow -a_{i1}L_1 + a_{11}L_i$$

$$\begin{aligned} i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\ &= -2(x + 2y - 3z - 6) + 1(2x - y + 4z - 2) \\ &= -2x - 4y + 6z + 12 + 2x - y + 4z - 2 \\ &= -5y + 10z + 10 \end{aligned}$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$\begin{aligned}
&= -4(x + 2y - 3z - 6) + 1(4x + 3y - 2z - 14) \\
&= -4x - 8y + 12z + 24 + 4x + 3y - 2z - 14 \\
&= -5y + 10z + 10
\end{aligned}$$

Thus we obtain the following system

$$\begin{aligned}
x + 2y - 3z &= 6 \\
-5y + 10z &= -10 \\
-5y + 10z &= -10
\end{aligned}$$

$ \begin{array}{cc} \begin{array}{c} a_{11} \\ \uparrow \end{array} & \begin{array}{c} a_{12} \\ \uparrow \end{array} \\ -5y + 10z = -10 \\ \begin{array}{c} a_{21} \\ \uparrow \end{array} & \begin{array}{c} a_{22} \\ \uparrow \end{array} \\ -5y + 10z = -10 \end{array} $
--

Again, 2nd time:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{aligned}
&= -(-5)(-5y + 10z + 10) + (-5)(-5y + 10z + 10) \\
&= -25y + 50z + 50 + 25y - 50z - 50 \\
&= 0
\end{aligned}$$

Thus we obtain the following system

$$\begin{aligned}
x + 2y - 3z &= 6 \\
-5y + 10z &= -10 \\
0 &= 0
\end{aligned}$$

(see step-04)

It occurs that,

$$0 \times x + 0 \times y + 0 \times z = 0$$

Hence we can delete the equation without affecting the solution

Thus we obtain the following new system

$$\begin{aligned}
x + 2y - 3z &= 6 \\
-5y + 10z &= -10
\end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2)=1 free variable, which is z.

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y, hence free variable is z]

$$\begin{aligned}
\text{Let, } z &= 1 \\
-5y + 10z &= -10 \\
-5y + 10 &= -10 \\
\Rightarrow -5y &= -20 \\
\therefore y &= 4
\end{aligned}$$

Again,

$$x + 2y - 3z = 6$$

$$x + 2 \times 4 - 3 \times 1 = 6$$

$$\Rightarrow x + 8 - 3 = 6$$

$$\Rightarrow x + 5 = 6$$

$$\Rightarrow x = 6 - 5$$

$$\therefore x = 1$$

Ans: $(x, y, z) = (1, 4, 1)$

Example 35: Solve the system

$$x + 2y - z = 0$$

$$2x + 5y + 2z = 0$$

$$x + 4y + 7z = 0$$

$$x + 3y + 3z = 0$$

Sol: We have,

1st time:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -2(x + 2y - z) + 1(2x + 5y + 2z)$$

$$= -2x - 4y + 2z + 2x + 5y + 2z$$

$$= y + 4z$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -1(x + 2y - z) + 1(x + 4y + 7z)$$

$$= -x - 2y + z + x + 4y + 7z$$

$$= 2y + 8z$$

$$i = 4, L_4 \rightarrow -a_{41}L_1 + a_{11}L_4$$

$$= -1(x + 2y - z) + 1(x + 3y + 3z)$$

$$= -x - 2y + z + x + 3y + 3z$$

$$= y + 4z$$

Then we obtain the system of equations

$$x + 2y - z = 0$$

$$y + 4z = 0$$

$$2y + 8z = 0$$

$$y + 4z = 0$$

Then we obtain the system of equations

$$x + 2y - z = 0$$

$$y + 4z = 0$$

$$y + 4z = 0$$

$$y + 4z = 0$$

[Dividing by 2]

Again, **2nd time:**

Since the 2nd, 3rd & 4th equations are identical. We can disregard one of them

$$\begin{aligned} x + 2y - z &= 0 \\ y + 4z &= 0 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z .

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y , hence free variable is z]

Let, $z = 1$,

$$y + 4z = 0$$

$$y = -4$$

$$x = -2y + z$$

$$\Rightarrow x = -2 \times (-4) + 1$$

$$\Rightarrow x = 8 + 1$$

$$\therefore x = 9$$

Ans: $(x, y, z) = (9, -4, 1)$

Example 36: Solve the system:

$$x + y - z = 0$$

$$2x + 4y - z = 0$$

$$3x + 2y + 2z = 0$$

Sol:

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

1st time:

$$i > 1$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -2(x + y - z) + 1(2x + 4y - z)$$

$$= -2x - 2y + 2z + 2x + 4y - z$$

$$= 2y + z$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -3(x + y - z) + 1(3x + 2y + 2z)$$

$$= -3x - 3y + 3z + 3x + 2y + 2z$$

$$= -y + 5z$$

Then we obtain the system of equations

$$x + y - z = 0$$

$$2y + z = 0$$

$$-y + 5z = 0$$

2nd time:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{array}{cc} a_{11} & a_{12} \\ \uparrow & \uparrow \\ 2y + 1z = 0 \\ \\ a_{21} & a_{22} \\ \uparrow & \uparrow \\ -1y + 5z = 0 \end{array}$$

$$\begin{aligned}
 &= -(-1)(2y + z) + 2(-y + 5z) \\
 &= 2y + z - 2y + 10z \\
 &= 11z
 \end{aligned}$$

Then we obtain the system of equations

$$x + y - z = 0$$

$$2y + z = 0$$

$$11z = 0$$

In echelon form there are three equations in three unknowns. Hence the system has a unique solution.

$$z = 0$$

$$y = 0$$

$$x = 0$$

$$\text{Ans : } (x, y, z) = (0, 0, 0)$$

Example 37: Solve the system,

$$2x + y - 2z + 3w = 1$$

$$3x + 2y - z + 2w = 4$$

$$3x + 3y + 3z - 3w = 5$$

Sol:

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

1st step:

$$i > 1$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -3(2x + y - 2z + 3w - 1) + 2(3x + 2y - z + 2w - 4)$$

$$= -6x - 3y + 6z - 9w + 3 + 6x + 4y - 2z + 4w - 8$$

$$= y + 4z - 5w - 5$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -3(2x + y - 2z + 3w - 1) + 2(3x + 3y + 3z - 3w - 5)$$

$$= -6x - 3y + 6z - 9w + 3 + 6x + 6y + 6z - 6w - 10$$

$$= 3y + 12z - 15w - 7$$

Then we obtain the system of equations

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5$$

$$3y + 12z - 15w = 7$$

2nd time:

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5 \text{ ----- } L_1$$

$$3y + 12z - 15w = 7 \text{ ----- } L_2$$

$\uparrow a_{11}$	$\uparrow a_{12}$	$\uparrow a_{13}$	
1	y	4	z - 5w = 5
$\uparrow a_{21}$	$\uparrow a_{22}$	$\uparrow a_{23}$	
3	y	12	z - 15w = 7

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$\begin{aligned} i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\ &= -3(y + 4z - 5w - 5) + 1(3y + 12z - 15w - 7) \\ &= -3y - 12z + 15w + 15 + 3y + 12z - 15w - 7 \\ &= 8 \end{aligned}$$

Then we obtain the new system of equations

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5$$

$$0 = -8$$

$$\text{Here, } 0 \times x + 0 \times y + 0 \times z + 0 \times w = -8$$

$$\Rightarrow 0 = -8$$

$$\text{But, } -8 \neq 0 \quad [\text{See Step-03}]$$

So, the system is inconsistent and has no solution.

Ans: The system has no solution.

Example 38: Solve the system,

$$x + y + z = 0$$

$$x + 3y + z = 0$$

$$x + y + z = 0$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{aligned} L_2 &\rightarrow -1(x + y + z) + 1(x + 3y + z) \\ &= 2y \end{aligned}$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$\begin{aligned} L_3 &\rightarrow -1(x + y + z) + 1(x + y + z) \\ &= 0 \end{aligned}$$

Thus we obtain the following system

$$x + y + z = 0$$

$$2y = 0$$

$$0 = 0$$

$$\Rightarrow x + y + z = 0$$

$$y = 0$$

We get, $y = 0$ then

$$\Rightarrow x + 0 + z = 0$$

Thus we obtain the following system

$$\Rightarrow \textcircled{x + z} = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is z .

Let,

$$z = 1$$

$$\therefore x = -1$$

$$\text{Ans: } (x, y, z) = (-1, 0, 1)$$

Example 39: Find the condition on λ and μ so that the following system of linear equations will have i) a unique solution ii) more than one solution iii) no solution

$$2x + 3y + z = 5$$

$$3x - y + \lambda z = 2$$

$$x + 7y - 6z = \mu$$

Solution: We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -3(2x + 3y + z - 5) + 2(3x - y + \lambda z - 2)$$

$$L_2 \rightarrow -6x - 9y - 3z + 15 + 6x - 2y + 2\lambda z - 4$$

$$L_2 \rightarrow -11y - 3z + 2\lambda z + 11$$

$$L_2 \rightarrow -11y + (2\lambda - 3)z + 11 \text{ -----(i)}$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -1(2x + 3y + z - 5) + 2(x + 7y - 6z - \mu)$$

$$L_3 \rightarrow -2x - 3y - z + 5 + 2x + 14y - 12z - 2\mu$$

$$L_3 \rightarrow 11y - 13z + 5 - 2\mu \text{ -----(ii)}$$

Thus we obtain the following system

$$2x + 3y + z = 5$$

$$-11y + (2\lambda - 3)z = -11 \text{ -----}L_1$$

$$11y - 13z = -5 + 2\mu \text{ -----}L_2$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -11\{-11y + (2\lambda - 3)z + 11\} - 11(11y - 13z + 5 - 2\mu)$$

$$L_2 \rightarrow 121y - 22\lambda z + 33z - 121 - 121y + 143z - 55 + 22\mu$$

$$L_2 \rightarrow -22\lambda z + 176z - 176 + 22\mu$$

$$L_2 \rightarrow (-22\lambda + 176)z - 176 + 22\mu$$

Thus we obtain the following system

$$\begin{aligned} 2x + 3y + z &= 5 \\ -11y + (2\lambda - 3)z &= -11 \\ (-22\lambda + 176)z &= 176 - 22\mu \end{aligned}$$

Thus we obtain the following system

$$\begin{aligned} 2x + 3y + z &= 5 \\ -11y + (2\lambda - 3)z &= -11 \\ (-\lambda + 8)z &= 8 - \mu \end{aligned}$$

Thus we obtain the following system

$$\begin{aligned} 2x + 3y + z &= 5 \\ 11y - (2\lambda - 3)z &= 11 \\ (\lambda - 8)z &= \mu - 8 \end{aligned}$$

The above system is in echelon form . Now we consider the following three cases.

- For a unique solution, the coefficient of z in the 3rd equation must be non-zero, that is, $\lambda \neq 8$ and μ have any value. Therefore, the given system has unique solution for $\lambda \neq 8$ and arbitrary values of μ
- If $\lambda = 8$ and $\mu = 8$, then the third equation of the above system vanishes and the system will be in echelon form having two equations in three unknowns, then the system has $(3-2) = 1$ free variable, which is z . Hence the given system has more than one solution for $\lambda = 8$ and $\mu = 8$.
- If $\lambda = 8$ and $\mu \neq 8$, then the third equation of the above system is of the form $0 = \mu - 8$ [where $\mu - 8 \neq 0$]; which is inconsistent. Thus the system has no solution for $\lambda = 8$ and $\mu \neq 8$.

Linear independence and linear dependence

A collection of vectors is either linearly independent or linearly dependent. The vectors $\{v_1, v_2, v_3, v_4, \dots, v_n\}$ are linearly independent if the equation involving linear combinations $a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 + \dots + a_nv_n = 0$ is true only when the scalars $a_i (= a_1, a_2, a_3, \dots, a_n)$ are all equal to zero. The vectors are linearly dependent if the equation has a solution when at least one of the scalars is not zero.

Example 40: Determine whether the following vectors in R^2 are linearly dependent or linearly independent.

$$x_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Solution: Suppose we have a linear combination of the vectors equal to 0

$$\begin{aligned}
 a_1x_1 + a_2x_2 + a_3x_3 &= 0 \\
 \Rightarrow a_1 \begin{bmatrix} -1 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} 5 \\ 6 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 4 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\
 \Rightarrow \begin{bmatrix} -a_1 \\ 3a_1 \end{bmatrix} + \begin{bmatrix} 5a_2 \\ 6a_2 \end{bmatrix} + \begin{bmatrix} a_3 \\ 4a_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\
 \Rightarrow \begin{bmatrix} -a_1 + 5a_2 + a_3 \\ 3a_1 + 6a_2 + 4a_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\
 \Rightarrow \begin{aligned} -a_1 + 5a_2 + a_3 &= 0 \\ 3a_1 + 6a_2 + 4a_3 &= 0 \end{aligned}
 \end{aligned}$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$\begin{aligned}
 L_i &\rightarrow -a_{i1}L_1 + a_{i1}L_i \\
 i = 2, L_2 &\rightarrow -a_{21}L_1 + a_{11}L_2 \\
 &= -3(-a_1 + 5a_2 + a_3) + (-1)(3a_1 + 6a_2 + 4a_3) \\
 &= 3a_1 - 15a_2 - 3a_3 - 3a_1 - 6a_2 - 4a_3 \\
 &= -21a_2 - 7a_3
 \end{aligned}$$

Thus we obtain the following new system

$$\begin{aligned}
 -a_1 + 5a_2 + a_3 &= 0 \\
 -21a_2 - 7a_3 &= 0
 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is a_3 .

Since, a_3 is free variable, we can choose it to be anything. In particular, we can choose it to be non-zero. Therefore, the vectors are linearly dependent.

Example 41: Determine whether the following vectors in R^3 are linearly dependent or linearly independent.

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad x_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Solution: Suppose we have a linear combination of the vectors equal to 0

$$\begin{aligned}
 a_1x_1 + a_2x_2 + a_3x_3 &= 0 \\
 \Rightarrow a_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 \Rightarrow \begin{bmatrix} a_1 \\ 2a_1 \\ 3a_1 \end{bmatrix} + \begin{bmatrix} -2a_2 \\ a_2 \\ 0a_2 \end{bmatrix} + \begin{bmatrix} a_3 \\ 0a_3 \\ a_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}
 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} a_1 - 2a_2 + a_3 \\ 2a_1 + a_2 + 0.a_3 \\ 3a_1 + 0.a_2 + a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a_1 - 2a_2 + a_3 = 0$$

$$\Rightarrow 2a_1 + a_2 + 0.a_3 = 0$$

$$3a_1 + 0.a_2 + a_3 = 0$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -2(a_1 - 2a_2 + a_3) + (1)(2a_1 + a_2)$$

$$= -2a_1 + 4a_2 - 2a_3 + 2a_1 + a_2$$

$$= 5a_2 - 2a_3$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$= -3(a_1 - 2a_2 + a_3) + (1)(3a_1 + a_3)$$

$$= -3a_1 + 6a_2 - 3a_3 + 3a_1 + a_3$$

$$= 6a_2 - 2a_3$$

Thus we obtain the following new system

$$a_1 - 2a_2 + a_3 = 0$$

$$5a_2 - 2a_3 = 0 \text{ -----} L_1$$

$$6a_2 - 2a_3 = 0 \text{ -----} L_2$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$= -6(5a_2 - 2a_3) + (5)(6a_2 - 2a_3)$$

$$= -30a_2 + 12a_3 + 30a_2 - 10a_3$$

$$= 2a_3$$

Thus we obtain the following new system

$$a_1 - 2a_2 + a_3 = 0$$

$$5a_2 - 2a_3 = 0$$

$$2a_3 = 0$$

$$\text{So, } a_3 = 0; a_2 = 0 \quad a_1 = 0$$

Therefore, the vectors are linearly independent.

Example 42: Determine whether the set of matrices

$$\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right\} \text{ is linearly independent.}$$

Solution: we have,

$$c_1 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + c_5 \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} c_1 + c_4 + c_5 & c_1 + c_2 + c_5 \\ c_3 + c_4 & c_2 + c_3 + c_4 + c_5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Which is equivalent to the homogeneous system of equations

$$c_1 + c_4 + c_5 = 0 \text{ -----} L_1$$

$$c_1 + c_2 + c_5 = 0 \text{ -----} L_2$$

$$c_3 + c_4 = 0 \text{ -----} L_3$$

$$c_2 + c_3 + c_4 + c_5 = 0 \text{ -----} L_4$$

$$c_1 + c_4 + c_5 = 0 \text{ -----} L_1$$

$$c_1 + c_2 + c_5 = 0 \text{ -----} L_2$$

$$0.c_1 + 0.c_2 + c_3 + c_4 = 0 \text{ -----} L_3$$

$$0.c_1 + c_2 + c_3 + c_4 + c_5 = 0 \text{ -----} L_4$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{21}L_2$$

$$L_2 \rightarrow -1(c_1 + c_4 + c_5) + 1(c_1 + c_2 + c_5)$$

$$L_2 \rightarrow -c_1 - c_4 - c_5 + c_1 + c_2 + c_5$$

$$L_2 \rightarrow -c_4 + c_2$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{31}L_3$$

$$L_3 \rightarrow -0(c_1 + c_4 + c_5) + 1(0.c_1 + 0.c_2 + c_3 + c_4)$$

$$L_3 \rightarrow 0 + c_3 + c_4$$

$$L_3 \rightarrow c_3 + c_4$$

$$i = 4, L_4 \rightarrow -a_{41}L_1 + a_{41}L_4$$

$$L_4 \rightarrow -0(c_1 + c_4 + c_5) + 1(0.c_1 + c_2 + c_3 + c_4 + c_5)$$

$$L_4 \rightarrow 0 + c_2 + c_3 + c_4 + c_5$$

$$L_4 \rightarrow c_2 + c_3 + c_4 + c_5$$

Thus we obtain the following new system

$$c_1 + c_4 + c_5 = 0$$

$$-c_4 + c_2 = 0$$

$$c_3 + c_4 = 0$$

$$c_2 + c_3 + c_4 + c_5 = 0$$

$$c_1 + c_4 + c_5 = 0$$

$$c_2 + c_3 + c_4 + c_5 = 0$$

$$-c_4 + c_2 = 0$$

$$c_3 + c_4 = 0$$

$$c_1 + c_4 + c_5 = 0$$

$$c_2 + c_3 + c_4 + c_5 = 0 \quad \text{-----} L_1$$

$$c_2 - c_4 = 0 \quad \text{-----} L_2$$

$$0.c_2 + c_3 + c_4 = 0 \quad \text{-----} L_3$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -1(c_2 + c_3 + c_4 + c_5) + 1(c_2 - c_4)$$

$$L_2 \rightarrow -c_2 - c_3 - c_4 - c_5 + c_2 - c_4$$

$$L_2 \rightarrow -c_3 - c_4 - c_5 - c_4$$

$$L_2 \rightarrow -c_3 - 2c_4 - c_5$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -0(c_2 + c_3 + c_4 + c_5) + 1(0.c_2 + c_3 + c_4)$$

$$L_3 \rightarrow 0 + c_3 + c_4$$

$$L_3 \rightarrow c_3 + c_4$$

Thus we obtain the following new system

$$c_1 + c_4 + c_5 = 0$$

$$c_2 + c_3 + c_4 + c_5 = 0$$

$$-c_3 - 2c_4 - c_5 = 0$$

$$c_3 + c_4 = 0$$

$$c_1 + c_4 + c_5 = 0$$

$$c_2 + c_3 + c_4 + c_5 = 0$$

$$c_3 + 2c_4 + c_5 = 0 \quad \text{-----} L_1$$

$$c_3 + c_4 = 0 \quad \text{-----} L_2$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -1(c_3 + 2c_4 + c_5) + 1(c_3 + c_4)$$

$$L_2 \rightarrow -c_3 - 2c_4 - c_5 + c_3 + c_4$$

$$L_2 \rightarrow -c_4 - c_5$$

Thus we obtain the following new system

$$c_1 + c_4 + c_5 = 0$$

$$c_2 + c_3 + c_4 + c_5 = 0$$

$$c_3 + 2c_4 + c_5 = 0$$

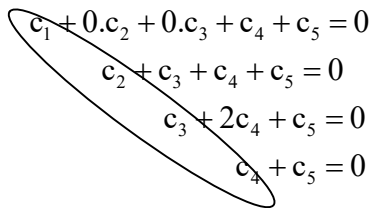
$$-c_4 - c_5 = 0$$

$$c_1 + c_4 + c_5 = 0$$

$$c_2 + c_3 + c_4 + c_5 = 0$$

$$c_3 + 2c_4 + c_5 = 0$$

$$c_4 + c_5 = 0$$



$$\begin{aligned} c_1 + 0.c_2 + 0.c_3 + c_4 + c_5 &= 0 \\ c_2 + c_3 + c_4 + c_5 &= 0 \\ c_3 + 2c_4 + c_5 &= 0 \\ c_4 + c_5 &= 0 \end{aligned}$$

In echelon form, there are only four equations in five unknowns, then the system has a non-zero solution and in particular $(5-4) = 1$ free variable, which is c_5 . Since, c_5 is free variable, we can choose it to be anything. That is, there are infinitely many solutions. In particular, we can choose it to be non-zero. Therefore, the set of vectors are linearly dependent.

Example 43: Determine whether the set $\{t^2 + 2t - 3, t^2 + 5t, 2t^2 - 4\}$ of vectors is linearly independent.

Solution: we have,

$$c_1(t^2 + 2t - 3) + c_2(t^2 + 5t) + c_3(2t^2 - 4) = 0$$

$$c_1t^2 + 2c_1t - 3c_1 + c_2t^2 + 5c_2t + 2c_3t^2 - 4c_3 = 0$$

$$(c_1 + c_2 + 2c_3)t^2 + (2c_1 + 5c_2)t + (-3c_1 - 4c_3) = 0$$

$$(c_1 + c_2 + 2c_3)t^2 + (2c_1 + 5c_2)t + (-3c_1 - 4c_3) = 0.t^2 + 0.t + 0 = 0$$

Taking coefficient of t^2 , t , on both sides

$$c_1 + c_2 + 2c_3 = 0$$

$$2c_1 + 5c_2 = 0$$

$$-3c_1 - 4c_3 = 0$$

$$c_1 + c_2 + 2c_3 = 0 \quad \text{-----} L_1$$

$$2c_1 + 5c_2 = 0 \quad \text{-----} L_2$$

$$-3c_1 - 4c_3 = 0 \quad \text{-----} L_3$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -2(c_1 + c_2 + 2c_3) + 1(2c_1 + 5c_2)$$

$$L_2 \rightarrow -2c_1 - 2c_2 - 4c_3 + 2c_1 + 5c_2$$

$$L_2 \rightarrow 3c_2 - 4c_3$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -(-3)(c_1 + c_2 + 2c_3) + 1(-3c_1 - 4c_3)$$

$$L_3 \rightarrow 3c_1 + 3c_2 + 6c_3 - 3c_1 - 4c_3$$

$$L_3 \rightarrow 3c_2 + 2c_3$$

Thus we obtain the following new system

$$c_1 + c_2 + 2c_3 = 0$$

$$3c_2 - 4c_3 = 0$$

$$3c_2 + 2c_3 = 0$$

$$c_1 + c_2 + 2c_3 = 0$$

$$3c_2 - 4c_3 = 0 \quad \text{-----} L_1$$

$$3c_2 + 2c_3 = 0 \quad \text{-----} L_2$$

To reduce it to a simpler system for each $i > 1$ applies the operation:

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -3(3c_2 - 4c_3) + 3(3c_2 + 2c_3)$$

$$L_2 \rightarrow -9c_2 + 12c_3 + 9c_2 + 6c_3$$

$$L_2 \rightarrow 18c_3$$

Thus we obtain the following new system

$$c_1 + c_2 + 2c_3 = 0$$

$$3c_2 - 4c_3 = 0$$

$$18c_3 = 0$$

$$\text{So, } c_3 = 0; c_2 = 0; c_1 = 0$$

Therefore, the vectors are linearly independent.

Home Task: Determine which of the following sets of vectors are linearly dependent.

$$1) \bar{a} = (1, 2) \quad , \quad \bar{b} = (3, 4)$$

$$2) \bar{c} = (2, 5) \quad , \quad \bar{d} = (4, 10)$$

$$3) \bar{e} = (1, 0), \quad \bar{f} = (2, 1) \quad \bar{g} = (3, 2)$$

$$4) \bar{h} = (1, 2, 1), \quad \bar{k} = (2, 3, 2)$$

Express vector $\bar{a} = (a_1, a_2)$ as a linear combination of vectors $(1, 0), (0, 1)$.

$\bar{a} = \lambda_1(1, 0) + \lambda_2(0, 1) = (a_1, a_2)$ thus $\lambda_1 = a_1$ and $\lambda_2 = a_2$ and therefore $\bar{a} = a_1(1, 0) + a_2(0, 1)$

In two-dimensional space, vectors $(1, 0)$ and $(0, 1)$ form a base, that means that any vector in this space can be expressed as a linear combination of $(1, 0)$ and $(0, 1)$. These vectors are usually denoted by $\underline{i} = (1, 0), \underline{j} = (0, 1)$

Express vector $\bar{b} = (b_1, b_2, b_3)$ as a linear combination of vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$.

In three-dimensional space, $\bar{b} = (b_1, b_2, b_3) = \lambda_1(1, 0, 0) + \lambda_2(0, 1, 0) + \lambda_3(0, 0, 1)$

Thus

$$b_1 = \lambda_1$$

$$b_2 = \lambda_2$$

$$b_3 = \lambda_3$$

and vector $\bar{b}$ can be expressed

$$\bar{b} = b_1(1, 0, 0) + b_2(0, 1, 0) + b_3(0, 0, 1)$$

In three-dimensional space, vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ form a base. Any vector in this space can be represented as a linear combination of these vectors.

They are denoted by

$$\underline{i} = (1, 0, 0)$$

$$\underline{j} = (0, 1, 0)$$

$$\underline{k} = (0, 0, 1)$$

Home Task

1. A manufacturer receives daily shipments of 70,000 springs and 45,000 pounds of stuffing for producing regular and support mattresses. Regular mattresses r requires 50 springs and 30 pounds of stuffing; support mattresses s requires 60 springs and 40 pounds of stuffing. How many mattresses of each type should be produced daily to use all available inventories?
2. A manufacturer produces desks and bookcases. Desks d require 5 hr of cutting time and 10 hr of assembling time. Bookcases b requires 15 min of cutting time and 1 hr of assembling time. Each day the manufacturer has available 200 personnel hours for cutting and 500 personnel hours for assembling. How many desks and bookcases should be scheduled for completion each day to use all available workpower?

Hints:

1.

$$50r + 60s = 70000$$

$$30r + 40s = 45000$$

2.

$$5d + 0.25b = 200$$

$$10d + b = 500$$

Polynomials of Matrices and Linear Operators

Consider a polynomial $f(t)$ over a field k

$$f(t) = a_n t^n + a_{n-1} t^{n-1} + a_{n-2} t^{n-2} + \dots + a_1 t + a_0$$

If A is a square matrix over k

Then we define

$$f(A) = a_n A^n + a_{n-1} A^{n-1} + a_{n-2} A^{n-2} + \dots + a_1 A + a_0 I$$

Example 44: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $f(t) = 2t^2 - 3t + 7$; $g(t) = t^2 - 5t - 2$ then $f(A)$ and $g(A)$.

Solution:

$$\text{Given, } f(t) = 2t^2 - 3t + 7$$

$$\therefore f(A) = 2A^2 - 3A + 7I$$

$$\Rightarrow f(A) = 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 3 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow f(A) = 2 \begin{bmatrix} 1 \times 1 + 2 \times 3 & 1 \times 2 + 2 \times 4 \\ 3 \times 1 + 4 \times 3 & 3 \times 2 + 4 \times 4 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\Rightarrow f(A) = 2 \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\Rightarrow f(A) = \begin{bmatrix} 14 & 20 \\ 30 & 44 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\Rightarrow f(A) = \begin{bmatrix} 18 & 14 \\ 21 & 39 \end{bmatrix}$$

Given, $g(t) = t^2 - 5t - 2$

$$\therefore g(A) = A^2 - 5A - 2I$$

$$\Rightarrow g(A) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 5 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow g(A) = \begin{bmatrix} 1 \times 1 + 2 \times 3 & 1 \times 2 + 2 \times 4 \\ 3 \times 1 + 4 \times 3 & 3 \times 2 + 4 \times 4 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\Rightarrow g(A) = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\Rightarrow g(A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Geometrical Significance of Eigen values and Eigen Vectors (Eigenfaces)

Suppose we have a matrix A: $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$

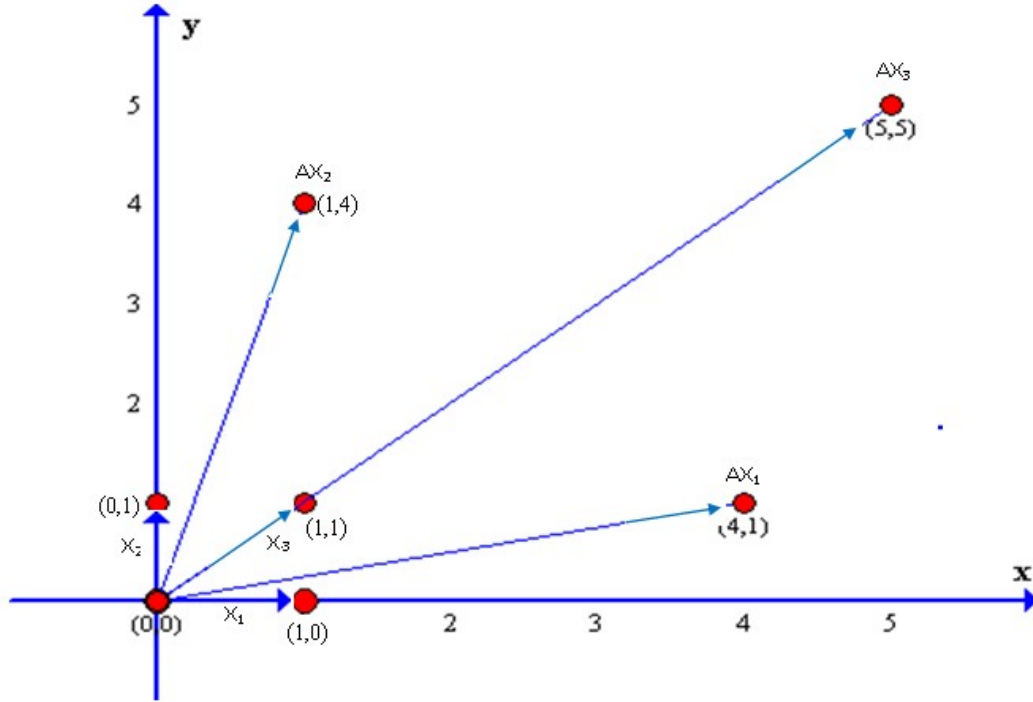


Figure 10

Suppose we take the product Ax_1 , where $x_1 = [1, 0] = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, as shown in Fig 10

$$\text{Then, } Ax_1 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \times 1 + 1 \times 0 \\ 1 \times 1 + 4 \times 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \text{-----(i)}$$

By comparing the vectors x_1 and Ax_1 we see that the product vector is scaled and rotated counter-clockwise with respect to x_1

Now consider the case where $x_2 = [0, 1] = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$,

$$\text{Then } Ax_2 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 0 + 1 \times 1 \\ 1 \times 0 + 4 \times 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} = [1, 4] \text{-----(ii)}$$

Here, we note a clockwise rotation of Ax_2 with respect to x_2

Now let's consider a more interesting case. Suppose $x_3 = [1, 1] = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$\text{Then } Ax_3 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 1 + 1 \times 1 \\ 1 \times 1 + 4 \times 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = [5, 5] \text{ -----(iii)}$$

Now the product vector points in the same direction as x_3 . The vector Ax_3 is scaled version of the vector x_3 . Because of this property, $x_3 = [1, 1]$ is an Eigen vector of A. The scale factor (which in this case is 5) is given the symbol λ and is referred to as an Eigen value.

$$\text{Thus } Ax_3 = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 1 + 1 \times 1 \\ 1 \times 1 + 4 \times 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow Ax_3 = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow Ax_3 = 5x_3 \because x_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ -----(iv)}$$

Here x_3 is an Eigen vector 5 is an Eigen value

In general we can say the relation of Eigen value and Eigen vector is $AX = \lambda X$

Summary of Eigen vectors (Eigenfaces) and Eigen values:

An Eigen vector is a nonzero vector that satisfies the equation $A\vec{V} = \lambda\vec{V}$, where A is a square matrix, λ is a scalar, and $\vec{v}$ is the eigenvector. λ is called an eigenvalue.

Or An Eigen vector is a nonzero vector that satisfies the equation $A\vec{X} = \lambda\vec{X}$, where A is a square matrix, λ is a scalar value, and $\vec{X}$ is the eigenvector. λ is called an eigenvalue

That is, the relation between Eigen-vectors and Eigen values of a matrix $A\vec{X} = \lambda\vec{X}$

Where A is given matrix, λ is Eigen values of the given matrix, $\vec{X}$ is corresponding Eigen vectors of the matrix

N.B: Eigenvalues are coefficients applied to eigenvectors that give the vectors their length or magnitude.

Example 45: Determine which of the following vectors are Eigen vectors for

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$ showing your analysis procedure graphically using graph paper.

(a) $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (b) $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ (c) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ (d) $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ (e) $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ (f) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Answer:

We have the relation between Eigen-vectors and Eigen values of a matrix $A \vec{X} = \lambda \vec{X}$

Where A is given matrix, λ is Eigen values of the given matrix, $\vec{X}$ is corresponding Eigen vectors of the matrix.

Test (a):

Given, $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 - 2 \times 1 \\ -4 \times 1 - 7 \times 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -11 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

$A \vec{X} = \lambda \vec{X}$ is not satisfied since $\vec{X}$ is not returned

Test (b):

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 3 \\ -4 \times 2 + 7 \times 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$$

$AX = \lambda X$ is not satisfied, since $\vec{X}$ is not returned

Test (c):

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 2 \\ -4 \times 1 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ A & X & \lambda & X \end{array}$$

$\therefore AX = \lambda X$ is satisfied; since $\vec{X}$ is returned

Test (d):

$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$, $X = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$$\begin{aligned}\therefore \mathbf{AX} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 2 \\ -4 \times 2 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \therefore \mathbf{AX} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ &\quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad \mathbf{A} \quad \mathbf{X} \quad \lambda \quad \mathbf{X} \\ \therefore \mathbf{AX} &= \lambda \mathbf{X} \text{ is satisfied, since } \vec{\mathbf{X}} \text{ is returned}\end{aligned}$$

Test (e):

$$\begin{aligned}\mathbf{A} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \\ \therefore \mathbf{AX} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 4 \\ -4 \times 2 + 7 \times 4 \end{pmatrix} = \begin{pmatrix} 10 \\ 20 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 4 \end{pmatrix} \\ \therefore \mathbf{AX} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 4 \end{pmatrix} \\ &\quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad \mathbf{A} \quad \mathbf{X} \quad \lambda \quad \mathbf{X}\end{aligned}$$

$\therefore \mathbf{AX} = \lambda \mathbf{X}$ is satisfied; since $\vec{\mathbf{X}}$ is returned

Test (f):

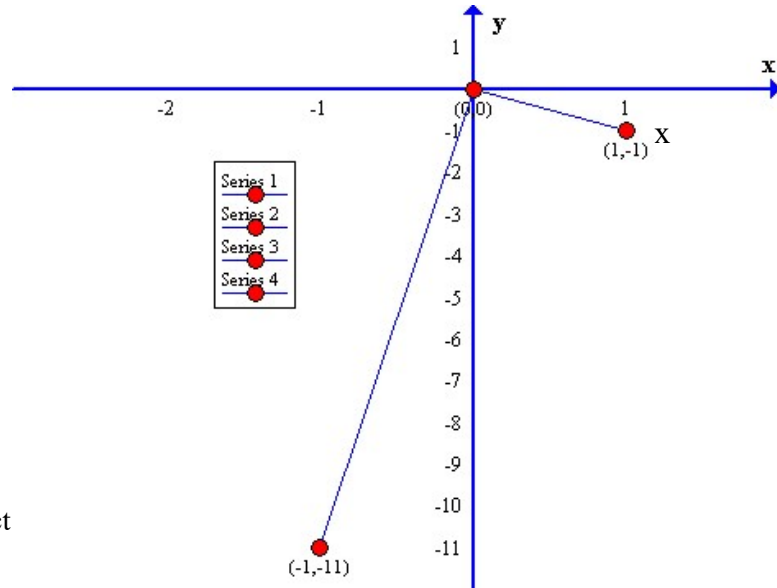
$$\begin{aligned}\mathbf{A} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ \therefore \mathbf{AX} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 1 \\ -4 \times 1 + 7 \times 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ \therefore \mathbf{AX} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &\quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad \mathbf{A} \quad \mathbf{X} \quad \lambda \quad \mathbf{X}\end{aligned}$$

$\therefore \mathbf{AX} = \lambda \mathbf{X}$ is satisfied; since $\vec{\mathbf{X}}$ is returned

Graphical Analysis:

Test (a):

$$\begin{aligned}\text{Given, } \mathbf{A} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, \mathbf{X} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ \therefore \mathbf{AX} &= \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 - 2 \times 1 \\ -4 \times 1 - 7 \times 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -11 \end{pmatrix}\end{aligned}$$



The product

Test (b):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 3 \\ -4 \times 2 + 7 \times 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$$

ctor of A

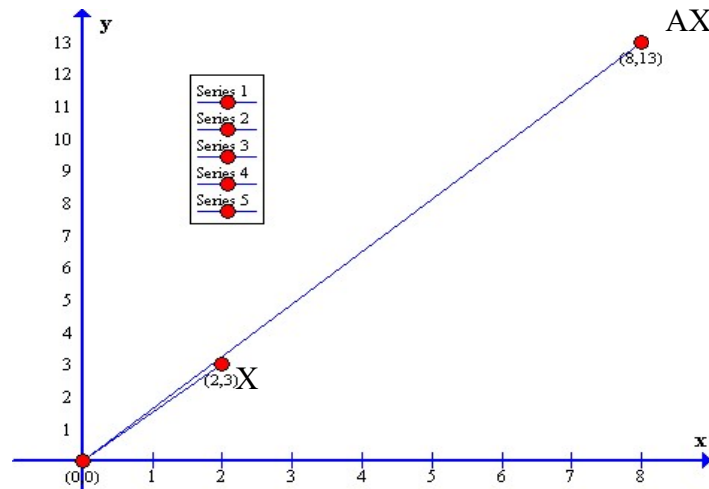


Figure no 12

The product vector AX is not in the same direction as X, So X is not eigen vector of A

Test (c):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 2 \\ -4 \times 1 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$$

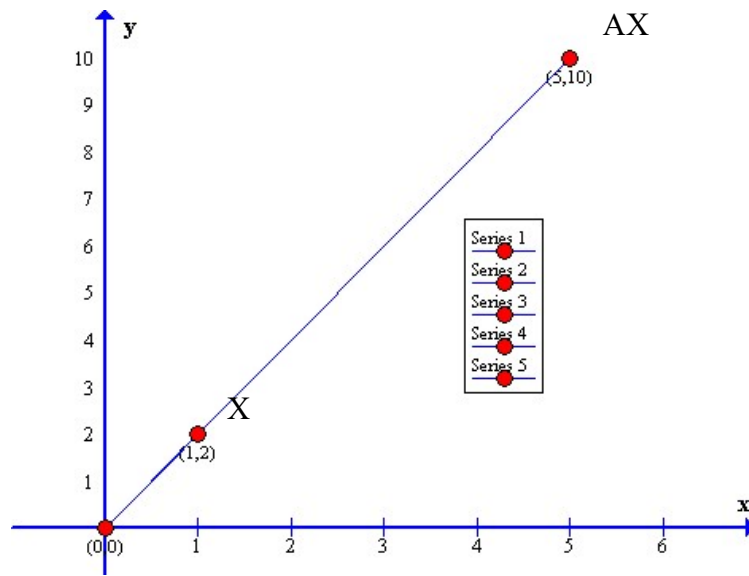


Figure no 13

The product vector AX is in the same direction as X, So X is an Eigen vector of A.

Test (d):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 2 \\ -4 \times 2 + 7 \times 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix}$$

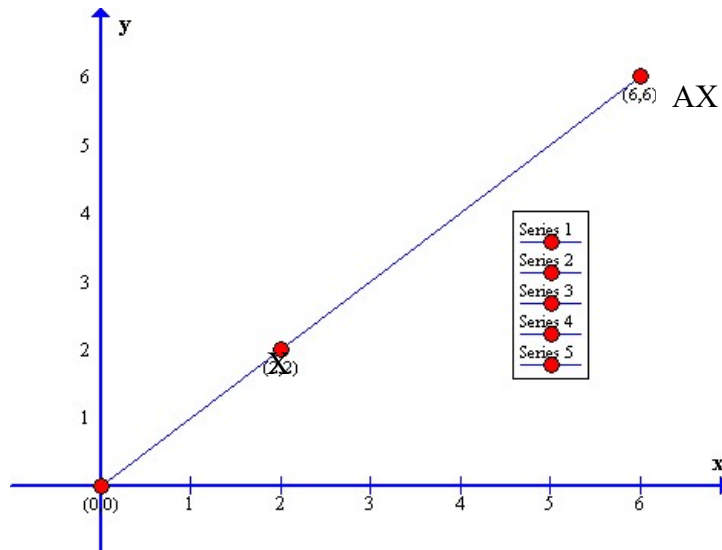


Figure no 14

The product vector AX is in the same direction as X, So X is an Eigen vector of A.

Test (e):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times 4 \\ -4 \times 2 + 7 \times 4 \end{pmatrix} = \begin{pmatrix} 10 \\ 20 \end{pmatrix}$$

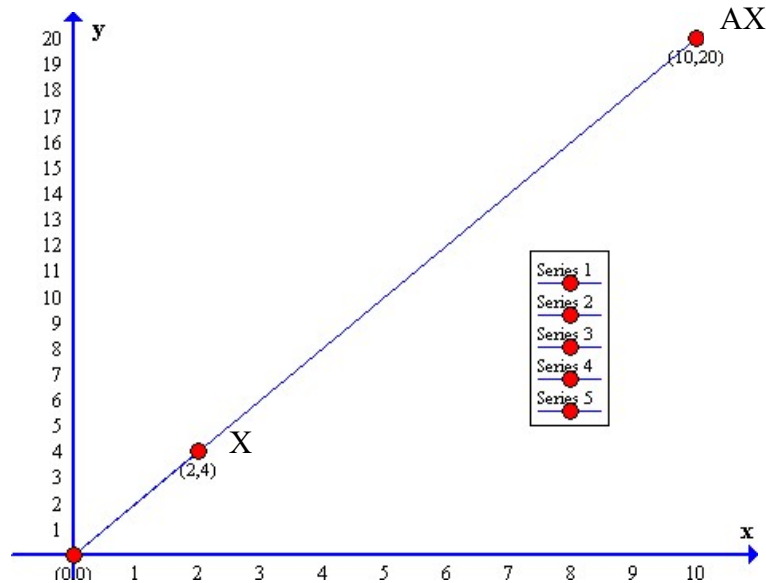


Figure no 15

The product vector AX is in the same direction as X, So X is an Eigen vector of A.

Test (f):

$$A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}, X = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 2 \times 1 \\ -4 \times 1 + 7 \times 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

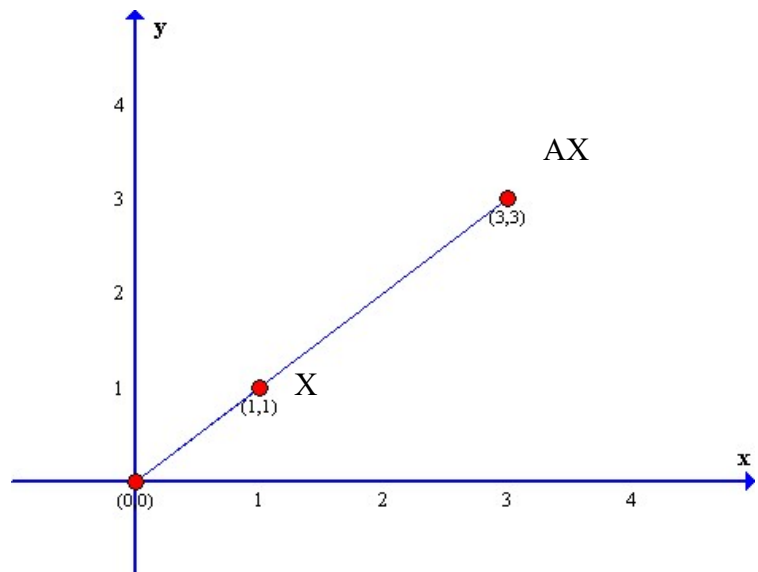


Figure no 16

The product vector AX is in the same direction as X, So X is an Eigen vector of A.

Three Dimensional Vector Representations

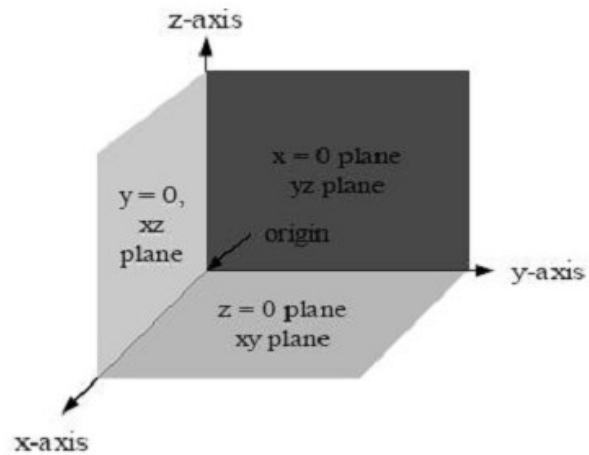


Figure 17

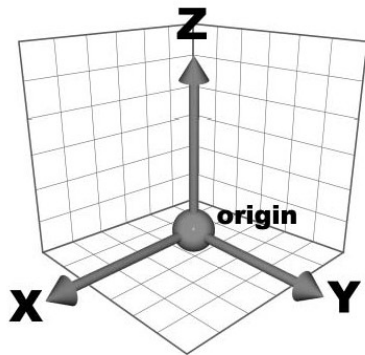


Figure no 18

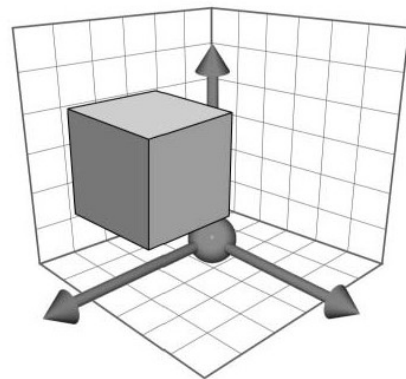


Figure no 19

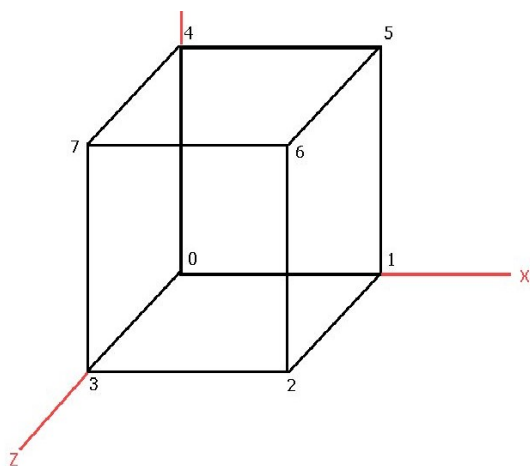


Figure no 20

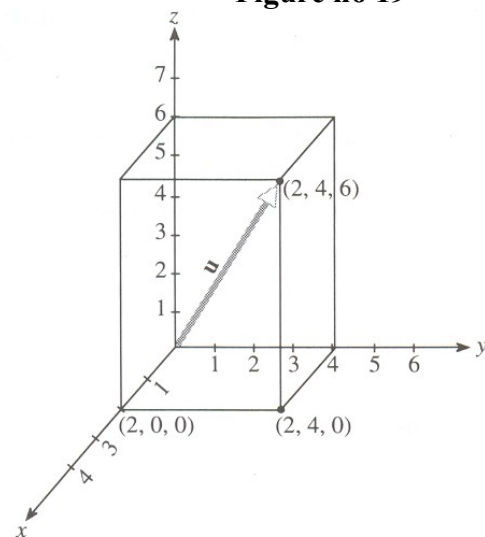


Figure no 21

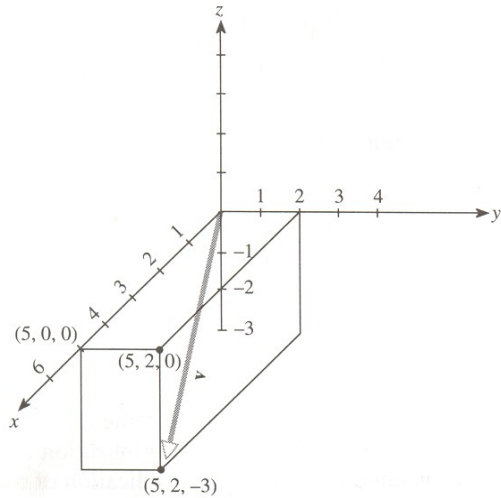


Figure 22

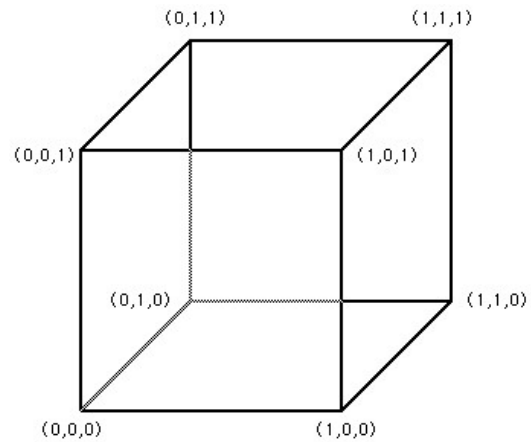


Figure 23

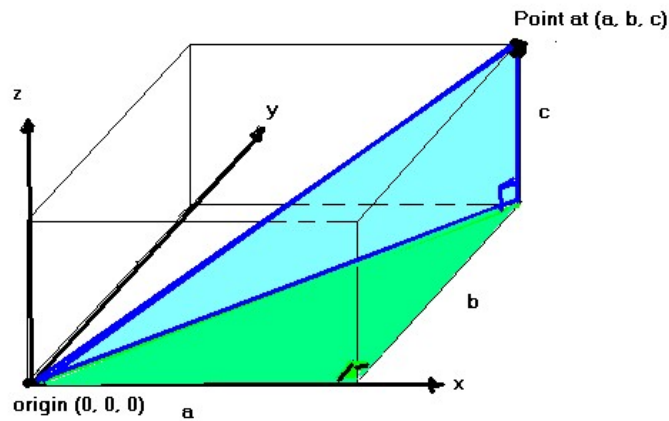


Figure no 24

Each axis of the cube represents values of red, green, or blue in the range [0,255].

- $(0, 0, 0)$ is black
- $(255, 255, 255)$ is white
- $(255, 0, 0)$ is red
- $(0, 255, 0)$ is green
- $(0, 0, 255)$ is blue
- $(255, 255, 0)$ is yellow
- $(0, 255, 255)$ is cyan
- $(255, 0, 255)$ is magenta

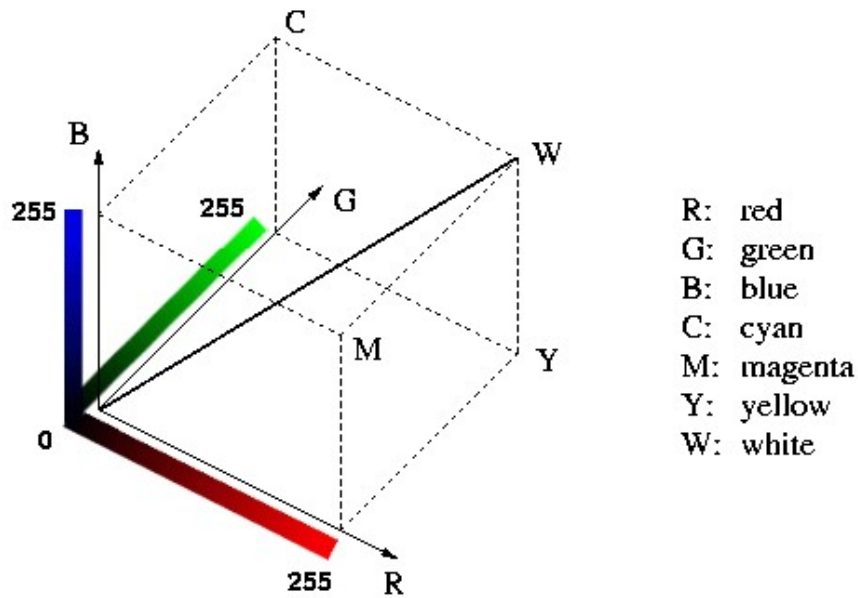


Figure 25: The RGB Color Cube

Note:

1. Each 8 bits represents the level of a primary colour - Red / Green / Blue (RGB)
2. Each pixel can be stored as a 24-bit number.
3. In hex it becomes obvious: - 2 digits / colour
4. $00FF00_{16}$ is the brightest *pure green*
5. Single hex digit = four bits
6. 2 hex digits per **byte**

Example 46: Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

Solution:

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{bmatrix}$$

$$\therefore |A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix}$$

$$\begin{aligned}
&= (1 - \lambda)(2 - \lambda) - 6 \\
&= 2 - \lambda - 2\lambda + \lambda^2 - 6 \\
&= \lambda^2 - 3\lambda - 4 \\
&= (\lambda^2 - 4\lambda + \lambda - 4)
\end{aligned}$$

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

$$\Rightarrow (\lambda^2 - 4\lambda + \lambda - 4) = 0$$

$$\Rightarrow (\lambda - 4)(\lambda + 1) = 0$$

$$\Rightarrow \lambda = 4 \text{ and } \lambda = -1$$

$$\text{Let a non-zero vector } \mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}$$

We have,

$$\begin{aligned}
&\mathbf{AX} = \lambda \mathbf{X} \\
&\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix} \\
\Rightarrow &\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = -1 \begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{When } \lambda = -1] \\
\Rightarrow &\begin{bmatrix} x + 2y \\ 3x + 2y \end{bmatrix} = -1 \begin{bmatrix} x \\ y \end{bmatrix} \\
\Rightarrow &\begin{bmatrix} x + 2y \\ 3x + 2y \end{bmatrix} = \begin{bmatrix} -x \\ -y \end{bmatrix}
\end{aligned}$$

Then we obtain the system of equations

$$x + 2y = -x$$

$$3x + 2y = -y$$

$$\Rightarrow 2x + 2y = 0$$

$$3x + 3y = 0$$

$$\Rightarrow x + y = 0 \quad [\text{Dividing by 2 on both sides}]$$

$$x + y = 0 \quad [\text{Dividing by 3 on both sides}]$$

$$x + y = 0 \quad \text{-----} L_1$$

$$x + y = 0 \quad \text{-----} L_2$$

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -(x+y) + 1(x+y)$$

$$L_2 \rightarrow -x - y + x + y$$

$$L_2 \rightarrow 0$$

Then we obtain the system of equations

$$x + y = 0$$

$$0 = 0$$

The Second equation can be deleted without affecting the solution

So we get,

$$x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1) = 1$ free variable, which is y .

[Note Here total variables are two, i.e x, y and blocked variable is x , hence free variable is y]

Let, $y = 2 \therefore x = -2$

$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ is a non-zero Eigen vector belonging to the Eigen value $\lambda = -1$

Again, for $\lambda = 4$

$$AX = \lambda X$$

$$\begin{aligned} & \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 4 \begin{bmatrix} x \\ y \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} x + 2y \\ 3x + 2y \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \end{bmatrix} \end{aligned}$$

Then we obtain the system of equations

$$x + 2y = 4x$$

$$3x + 2y = 4y$$

$$\Rightarrow -3x + 2y = 0$$

$$3x - 2y = 0$$

$$\Rightarrow 3x - 2y = 0 \quad \text{[Multiplying by -1 on both sides]}$$

$$3x - 2y = 0$$

Since the 1st and 2nd equations are identical. We can disregard one of them

So, $3x - 2y = 0$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1) = 1$ free variable, which is y .

[Note Here total variables are two, i.e x, y and blocked variable is x , hence free variable is y]

Let, $y = 3$

$3x = 6 \therefore x = 2$

$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is a non-zero eigen vector belonging to the eigen value $\lambda = 4$.

$\therefore -1, 4$ are the eigen values of the matrix A and their respective eigen vectors are $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

Let P be the matrix whose columns are the Eigen vectors of the given matrix

$\therefore P = \begin{bmatrix} -2 & 2 \\ 2 & 3 \end{bmatrix}$

Example 47: Find the Eigen vectos of the matrix $\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$

Let, $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix}$

$A - \lambda I = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A - \lambda I = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

$A - \lambda I = \begin{bmatrix} 3-\lambda & 1 & 1 \\ 1 & 5-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{bmatrix}$

$\therefore |A - \lambda I| = \begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 5-\lambda & 1 \\ 1 & 1 & 3-\lambda \end{vmatrix}$

$= (3-\lambda)[(5-\lambda)(3-\lambda)-1] - 1[1(3-\lambda)-1] + 1(1-(5-\lambda))$

$$= (3 - \lambda)(\lambda^2 - 8\lambda + 15 - 1) - 1(3 - \lambda - 1) + 1(1 - 5 + \lambda)$$

$$= (3 - \lambda)(\lambda^2 - 8\lambda + 14) - (2 - \lambda) + (\lambda - 4)$$

$$= 3\lambda^2 - 24\lambda + 42 - \lambda^3 + 8\lambda^2 - 14\lambda - 2 + \lambda + \lambda - 4$$

$$= -\lambda^3 + 11\lambda^2 - 36\lambda + 36$$

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

$$-\lambda^3 + 11\lambda^2 - 36\lambda + 36 = 0$$

$$\Rightarrow \lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0 \text{ -----(i)}$$

If $\lambda = 3$ then the equation (i) is satisfied.

So $\lambda - 3$ is a factor of $\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$

Now, Follow the Remainder System

$$\begin{array}{r|l} \lambda - 3 & \lambda^3 - 11\lambda^2 + 36\lambda - 36 \\ & \lambda^3 - 3\lambda^2 \\ & \hline & -8\lambda^2 + 36\lambda - 36 \\ & -8\lambda^2 + 24\lambda \\ & \hline & +12\lambda - 36 \\ & +12\lambda - 36 \\ & \hline & -0 \end{array} \quad \lambda^2 - 8\lambda + 12$$

$$\Rightarrow \lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 8\lambda + 12) = 0 \text{(ii)}$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 6\lambda - 2\lambda + 12) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 6)(\lambda - 2) = 0$$

$$\therefore \lambda = 3, 6, 2$$

$$\text{Let a non-zero vector, } \mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We have,

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

$$\therefore \mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 3 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ [For } \lambda = 3\text{]} \\ \Rightarrow \begin{pmatrix} 3x + y + z \\ x + 5y + z \\ x + y + 3z \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \\ 3z \end{pmatrix}$$

Thus we obtain following system:

$$\begin{aligned} 3x + y + z &= 3x \\ x + 5y + z &= 3y \\ x + y + 3z &= 3z \end{aligned}$$

$$\Rightarrow \begin{aligned} y + z &= 0 \\ x + 2y + z &= 0 \\ x + y &= 0 \end{aligned}$$

Thus we obtain following system:

$$\begin{aligned} x + 2y + z &= 0 \\ x + y + 0.z &= 0 \\ y + z &= 0 \end{aligned}$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\begin{aligned} &= -1(x + 2y + z) + 1(x + y + 0.z) \\ &= -y - z \end{aligned}$$

Thus we obtain following system:

$$\begin{aligned} \therefore x + 2y + z &= 0 \\ -y - z &= 0 \\ y + z &= 0 \end{aligned}$$

Thus we obtain following system:

$$\begin{aligned} \therefore x + 2y + z &= 0 \\ y + z &= 0 & \text{[Multiplying by -1 on both sides]} \\ y + z &= 0 \end{aligned}$$

Since the 2nd and 3rd equations are identical. We can disregard one of them

Thus we obtain following system:

$$\begin{aligned} x + 2y + z &= 0 \\ y + z &= 0 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2)=1 free variable, which is z.

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y, hence free variable is z]

Let $z = 1$

$$\therefore y = -1$$

$$x = 1$$

$$v = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \text{ is a non-zero eigen vector belonging to the eigen value } \lambda = 3$$

Again

$$\text{Let a non-zero vector, } v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

We have,

$$Av = \lambda v$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 6 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ [For } \lambda = 6]$$

$$\Rightarrow \begin{pmatrix} 3x + y + z \\ x + 5y + z \\ x + y + 3z \end{pmatrix} = \begin{pmatrix} 6x \\ 6y \\ 6z \end{pmatrix}$$

$$\Rightarrow 3x + y + z = 6x$$

$$x + 5y + z = 6y$$

$$x + y + 3z = 6z$$

$$\Rightarrow -3x + y + z = 0$$

$$x - y + z = 0$$

$$x + y - 3z = 0$$

$$\Rightarrow 3x - y - z = 0$$

$$x - y + z = 0$$

$$x + y - 3z = 0$$

Thus we obtain the following system

$$3x - y - z = 0$$

$$x - y + z = 0$$

$$x + y - 3z = 0$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2, L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$L_2 \rightarrow -1(3x - y - z) + 3(x - y + z)$$

$$= -3x + y + z + 3x - 3y + 3z$$

$$= -2y + 4z$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -1(3x - y - z) + 3(x + y - 3z)$$

$$= -3x + y + z + 3x + 3y - 9z = 0$$

$$\Rightarrow 4y - 8z = 0$$

Thus we obtain following system:

$$3x - y - z = 0$$

$$-2y + 4z = 0$$

$$4y - 8z = 0$$

$$\Rightarrow 3x - y - z = 0$$

$$y - 2z = 0$$

$$y - 2z = 0$$

Since the 2nd and 3rd equations are identical. We can disregard one of them

$$\begin{array}{l} 3x - y - z = 0 \\ y - 2z = 0 \end{array}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable, which is z .

[Note Here total variables are three, i.e x, y, z and blocked variable is x and y , hence free variable is z]

Let,

$$z = 1$$

$$\therefore y = 2$$

$$x = 1$$

$\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ is a non-zero eigen vector belonging to the eigen value $\lambda = 6$

Again

Let a non-zero vector, $\mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

We have,

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

$$\therefore \mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad [\text{For } \lambda = 2]$$

$$\Rightarrow \begin{pmatrix} 3x + y + z \\ x + 5y + z \\ x + y + 3z \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix}$$

$$\Rightarrow 3x + y + z = 2x$$

$$x + 5y + z = 2y$$

$$x + y + 3z = 2z$$

$$\Rightarrow x + y + z = 0$$

$$x + 3y + z = 0$$

$$x + y + z = 0$$

We have,

$$\mathbf{L}_i \rightarrow -\mathbf{a}_{i1}\mathbf{L}_1 + \mathbf{a}_{i1}\mathbf{L}_i$$

$$i = 2, \mathbf{L}_2 \rightarrow -\mathbf{a}_{21}\mathbf{L}_1 + \mathbf{a}_{11}\mathbf{L}_2$$

$$L_2 \rightarrow -1(x+y+z) + 1(x+3y+z) \\ = 2y$$

$$i = 3, L_3 \rightarrow -a_{31}L_1 + a_{11}L_3$$

$$L_3 \rightarrow -1(x+y+z) + 1(x+y+z) \\ = 0$$

Thus we obtain the following system

$$x + y + z = 0$$

$$2y = 0$$

$$0 = 0$$

$$\Rightarrow \begin{matrix} x + y + z = 0 \\ y = 0 \end{matrix}$$

We get, $y = 0$ then

$$x + 0 + z = 0$$

Thus we obtain the following system

$$\Rightarrow x + z = 0$$

In echelon form, there is only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is z .

Let,

$$z = 1$$

$$\therefore x = -1$$

$$v = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ is a non-zero eigen vector belonging to the eigen value } \lambda = 2$$

Let P be the matrix whose columns are the Eigen vectors of the given matrix

$$P = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

Diagonalization: Let $T: v \rightarrow v$ be a linear operator on a vector space v with finite dimension n . Note that T can be represented by a diagonal matrix.

$$\begin{bmatrix} k_1 & 0 & 0 & 0 & - & - & 0 \\ 0 & k_2 & 0 & 0 & - & - & 0 \\ 0 & 0 & k_3 & 0 & - & - & 0 \\ 0 & 0 & 0 & k_4 & - & - & 0 \\ - & - & - & - & - & - & - \\ - & - & - & - & - & - & - \\ 0 & 0 & 0 & 0 & - & - & k_n \end{bmatrix}$$

If and only if there exists a basis $(v_1, v_2, v_3, \dots, v_n)$ of V for which

$$T(v_1) = k_1 v_1$$

$$T(v_2) = k_2 v_2$$

$$T(v_3) = k_3 v_3$$

.....

.....

$$T(v_n) = k_n v_n$$

That is, such that the vectors $v_1, v_2, v_3, \dots, v_n$ are eigen vectors of T belonging respectively to eigen values $k_1, k_2, k_3, \dots, k_n$.

Example 48: If D be the different operator on the vector space V of differentiable functions. Then $D(e^{a_k t}) = a_k e^{a_k t}$ accordingly $e^{a_1 t}, e^{a_2 t}, \dots, e^{a_n t}$ are Eigen vectors of D belonging to the distinct Eigen values $a_1, a_2, a_3, \dots, a_n$